Chapter 14

Lung and pleura

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Lung cancer pathology

Key facts

- Lung cancer is divided into ***small-cell lung cancer*** (SCLC) (25%) and ***non-small-cell lung cancer*** (NSCLC) (75%): surgery is rarely indicated for SCLC, which is generally treated with systemic chemotherapy, while NSCLC in amendable to surgery as part of stage-specific therapy.
  - SCLC includes ***oat cell*** and ***intermediate*** types.
  - NSCLC includes ***squamous cell*** (30%), ***adenocarcinoma*** (35%), ***bronchopulmonary neuroendocrine*** tumors (BPNET), and other tumor types, e.g., ***bronchoalveolar***, ***adenosquamous,*** and ***mucoepidermoid*** and ***adenoid cystic*** tumor (5%).
  - BPNET includes ***small-cell***, ***large-cell*** (10%), and ***carcinoid*** tumors
- 1 200 000 cases diagnosed annually worldwide. In the USA there are 170 000 new cases with 160 000 deaths per year: it is the number one cancer killer of both men and women. By 2025, 80% of all lung cancers will be in Asia.
- More lung cancer deaths than colon, breast, and prostate cancer combined.
- There is a rising incidence of adenocarcinoma, and a rising incidence of lung cancer in never-smokers.

Etiology

- 87% of lung cancers are due to smoking: 90% in men and 85% in women. A smoker is defined as someone who has smoked ≥100 cigarettes in their lifetime.
- 3000 lung cancer deaths/year due to second-hand smoke.
- The lifetime risk of cancer in smokers is 17% in men and 11.6% in women: if an individual stops smoking, their chance of developing lung cancer decreases.

Pathology of small-cell lung cancer

- Bronchopulmonary neuroendocrine tumors are often centrally located.
- Considered a systemic disease, with very little role for surgery:
  - Often metastatic at the time of surgery.
  - Treated with systemic chemotherapy.
- Characterized histologically by cells with scant cytoplasm in fibrous stoma, and abundant neurosecretory granules.

Pathology of non-small-cell lung cancer

NSCLC: adenocarcinoma (35%)

- Most common lung cancer.
- Generally caused by smoking or other tobacco products.
  - New data suggests the presence of epidermal growth factor receptor mutations in never-smokers.
  - Young age, Asian, female.
- Often peripherally located, or associated with scar and fibrosis, especially in COPD (so-called *scar carcinoma*).
- Histology: acinar or glandular patterns.

NSCLC: squamous cell carcinoma (30%)
- Generally caused by smoking or other tobacco products.
- 2/3 are centrally located.
- They can obstruct airways, causing postobstructive pneumonia.
- 10–20% develop central necrosis and cavitation.
- Histology: intercellular bridge formation and keratinization.

NSCLC: bronchioaveolar carcinoma (<5%)

Adenocarcinoma *in situ* (defined by the World Health Organization).
- Occurs in non-smokers.
- Three forms:
  - Solitary (ground glass opacity).
  - Multinodular.
  - Diffuse or pneumotic.
- Only rare direct invasion or lymph node metastasis.
- Histology: spread along the alveolus without direct invasion.

NSCLC: BPNET
- Neuroendocrine origin, derived from Kulchitsky cells, with amine precursor uptake and decarboxylation.
- Three common types: *large-cell*, *carcinoid*, and *small-cell lung cancer* which is classified separately (📖 p668).

NSCLC: large-cell carcinoma (10%)
- Bronchopulmonary neuroendocrine tumor common in the periphery.
- Two subtypes:
  - Giant cell.
  - Clear cell.
- Tends to have a very aggressive clinical presentation.
- Tumors may have cavitation.
- Histology: enlarged hyperchromatic nuclei with prominent nucleoli.

NSCLC: carcinoid tumor (1%)
- BPNET often centrally located.
- Not associated with smoking.
- Carcinoid syndrome is seldom seen (<1%).
- Two subtypes:
  - *Typical:* often endobronchial, rare to have vascular invasion, 10% incidence of nodal or distant metastasis, infrequent cellular mitosis, 90% 5-year survival
  - *Atypical:* often peripheral, frequent cellular mitosis, frequent vascular invasion, 50% incidence of nodal or distant metastasis, 25–69% 5-year survival.

Metastatic disease
- 40–50% of patients with newly diagnosed lung cancer have metastatic disease at time of diagnosis.
- Modes of spread: direct invasion, lymphatics, hematogenous.
- More common sites of spread include liver, adrenal, brain, and bone. Others: skin, heart, thyroid, small intestine, ovary, spleen, kidney, parathyroid, pituitary.

Diagnosis of lung cancer

Key facts

- 95% of patients are symptomatic at diagnosis, most are male >60 years.
- Symptoms may be local, regional, or systemic (see Box 14.1).
- Chest CT, PET, brain MRI, bronchoscopy, and usually needle biopsy are necessary for diagnosis and staging, with mediastinal staging by mediastinoscopy or less invasive modalities, e.g., EBUS.

Presentation

- ***Proximal tumors*** tend to produce symptoms of major airway obstruction and irritation (hemoptysis, dyspnea, cough, wheezing, stridor, hoarseness, Horner's syndrome, SVC syndrome, postobstructive pneumonia, pleural effusion, ***Pancoast syndrome***).
- ***Peripheral tumors*** are often asymptomatic or present with signs and symptoms of pleural or chest wall invasion or pleural effusion (10%) (pleuritic chest wall pain, progressive dyspnea).
- May present with symptoms and signs of ***metastatic spread***: neurological—headache, blurred vision, nausea, diplopia, decreased consciousness, ataxia; bony pain and pathological fracture; liver and abdominal pain, anorexia, jaundice, ascites, liver failure, hepatomegaly; adrenal glands—symptoms of Addison's disease.
- ***Paraneoplastic syndromes.***
- Always ask about history of tobacco smoking (number of pack years), asbestos exposure, employment history, unintentional weight loss >5% of body weight, and recent onset of joint pains, change in voice, chest wall pain, back pain, palpable LN.
- Examine for clubbing, palpate all lymph nodes, cutaneous lesions, pleural effusion, localize chest wall pain.

Paraneoplastic syndromes

10–20% of cases. Usually caused by SCLC and squamous cell carcinomas.

- ***Metabolic:*** Cushing syndrome, hypercalcemia, SIADH (see 📖 p671), carcinoid syndrome, gynecomastia, hyperthyroidism, hypoglycemia, excess growth hormone, excess prolactin, FSH, LH.
- ***Neurologic:*** encephalopathy, Lambert–Eaton syndrome, subacute cerebellar degeneration, peripheral neuropathy, polymyositis, autonomic neuropathy, myoclonus, optic neuritis.
- ***Skeletal:*** clubbing, HPOA (see 📖 p669).
- ***Hematological:*** anemia, thrombocytosis, thrombocytopenia, eosinophilia, red cell aplasia, leukoerythroblastosis, DIC, leukemoid reactions, migratory thrombophlebitis.
- ***Cutaneous:*** hyperkeratosis, dermatomyositis, acanthosis nigricans, hyperpigmentation, erythema gyratum repens, hypertrichosis lanuginose acquisita.
- ***Other:*** nephrotic syndrome, anorexia-cachexia, hypouricemia, secretion of VIP with diarrhea, hyperamylasemia.

Hypertrophic pulmonary osteoarthropathy (HPOA)
Periostitis of distal long bones, commonly tibia, fibula, and radius. Joint tenderness and swelling. It may involve metacarpals and metatarsals causing acute polyarthritis. Associated with clubbing. More common with adenocarcinoma. Etiology unknown.

Syndrome of inappropriate anti-diuretic hormone secretion (SIADH)
Associated with SCLC. Symptoms uncommon. Excess ADH secreted. Leads to hyponatremia, nausea and vomiting, anorexia, confusion, seizures, lethargy. Diagnosis made by low serum osmolality, high urinary excretion of sodium, and high urine osmolality.

Hypercalcemia
10% of lung cancers. ~15% of these are secondary to parathyroid hormone secretion. Most often associated with squamous cell carcinoma.

Ectopic adrenocorticotrophic syndrome
Usually occurs with SCLC. ACTH production is autonomous. <5% present with Cushing syndrome. Usual presentation is due to severe hypokalemia, hyperglycemia and metabolic alkalosis.

Neurologic paraneoplastic syndromes
Central neuropathies (brainstem encephalitis, encephalomyelitis, dementia) and many peripheral neuropathies (sensory, autonomic, and sensorimotor). Usual with small-cell lung cancer and occasionally with squamous cell carcinoma.

Peripheral neuromyopathy
The most common paraneoplastic syndrome related to lung cancer.

Box 14.1 Localizing symptoms of lung cancer

- Bronchopulmonary symptoms:
  - Cough.
  - Hemoptysis.
  - Chest pain.
  - Dyspnea.
  - Wheezing.
- Intrathoracic symptoms:
  - Pleural effusion (10%).
  - Hoarseness (recurrent laryngeal nerve).
  - SVC syndrome.
  - Chest wall pain.
  - Arm pain (brachial plexus).
  - Horner's syndrome.
  - Dysphagia.
  - Phrenic nerve paralysis.
- Paraneoplastic syndromes (2%).
- Symptoms from distant metastatic spread.

Investigation of lung cancer

Key facts

- Imaging is performed for diagnosis and staging.
- Proper staging of lung cancer includes chest CT, PET, bronchoscopy, brain MRI, and mediastinal staging.

PA and lateral chest radiograph

Allow rough localization of tumor and detection of gross hilar and mediastinal lymphadenopathy, pleural effusions, areas of consolidation, evidence of bony destruction of ribs and vertebral bodies, and elevation of the hemidiaphragm.

CT scan

- Helps define locoregional spread, invasion of local structures, synchronous lesions, and is the gold standard for determining T status.
- Allows accurate localization of primary lesion, allowing size measurements, assessment for evidence of local invasion.
- Provides detailed information regarding remaining lung parenchyma (presence of bullae, emphysema, interstitial lung disease, undiagnosed masses, pleural thickening, small pleural effusions).
- Extension into the upper abdomen should routinely be requested to allow assessment of liver and adrenal glands.
- Provides detailed information on intrathoracic lymph nodes. A positive result is one where a lymph node is >1cm in long axis.

PET scan

- Agent used is a labelled D-glucose analogue (fluorodeoxyglucose – FDG). Lesions >5mm diameter detected. False positives with granulomatous and inflammatory lesions. Useful in diagnosing mediastinal and distant metastatic disease.
- ***PET/CT:*** simultaneous PET scan and CT scan allowing more accurate localization of lesions including metastases when compared to either investigation alone.

MRI

- Brain MRI obtained if neurological symptoms, large tumors, or tumors with high metabolic activity on PET.
- Allows accurate assessment of spinal canal, bony lesions, and vascular structures.

Bronchoscopy

- Technique is described on 📖 pp654–657.
- Excellent modality for visualization and localization of endobronchial tumor, assessment of carina and rest of tracheobronchial tree, ***direct biopsy, bronchial washings (bronchoalveolar lavage)***, and ***sputum cytology***.
- Sensitivity of sputum cytology dependent on tumor size, location, and histological type. Central squamous cell tumors have highest yield. False-positive findings in up to 3%.

- Also allows ***EBUS-TBNA*** of lesions outside the tracheobronchial tree—subcarinal and paratracheal nodes.
- EBUS-TBNA has 69% sensitivity, 88% negative predictive value.
- A valuable tool when assessing preoperatively the extent of resection required for curative surgery as it identifies intrabronchial involvement, and possible need for bronchoplastic procedure.
- Often performed/repeated at the time of surgery.

Mediastinal staging

Mediastinoscopy

- Mediastinoscopy is the gold standard assessment of the mediastinum as it allows direct biopsy and histological assessment of mediastinal lymph nodes, hence determines presence or absence of N2/N3 disease.
- ***Cervical mediastinoscopy*** (📖 p660) for paratracheal lymph nodes.
- ***Anterior mediastinotomy*** (📖 pp662–663) (Chamberlain) or left VATS (📖 p686) for AP window station 5 and 6 lymph nodes.
- 2% false negative, 0.6% morbidity, 0.2% mortality.[1]

Thoracotomy (VATS)

Occasionally necessary for diagnosis and staging (<5% of cases). Often unable to distinguish T3 from T4 lesion prior to thoracotomy.

Reference

1. Hammoud Z *et al.* The current role of mediastinoscopy in the evaluation of thoracic disease. *J Thorac Cardiovasc Surg* 1999; **118**:894–9.

Staging of lung cancer

Key facts

- The TNM classification is used: 7th edition of the AJCC/IUAC cancer staging manual (2010) (see Table 14.1).
- The TNM class is based on ***tumor*** size and location, ***node*** size, and presence of ***metastases*** (see Table 14.2).
- Regional lymph nodes (Table 14.3 and Fig. 14.1) include double digit N1 (ipsilateral) or N3 (contralateral) nodes that are outside the pleural envelope, and single digit mediastinal N2 (ipsilateral) or N3 (contralateral).

Table 14.1 TNM classification for lung cancer

Primary tumor (T)	
Tx	**Primary tumor cannot be assessed**
T0:	**No evidence of primary tumor**
Tis:	**Carcinoma *in situ***
T1a:	≤2cm
T1b:	>2 , ≤3cm
T2a:	>3cm, ≤5cm
T2b:	>5cm, ≤ 7cm
T3:	>7cm; chest wall, diaphragm, phrenic nerve, mediastinal pleura, or parietal pericardial invasion; same lobe synchronous nodules; tumor in the main bronchus and <2cm distal to the carina; associated atelectasis or obstructive pneumonitis of the entire lung
T4:	Extension into mediastinal structures (e.g., heart, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, carina; ipsilateral lung synchronous nodules, but different lobes
Regional lymph nodes (N)	
Nx:	Regional lymph nodes cannot be assessed
N0:	No regional lymph node metastasis
N1:	Ipsilateral peribronchial, ipsilateral hilar, intrapulmonary lymph nodes
N2:	Ipsilateral mediastinal, subcarinal lymph nodes
N3:	Contralateral mediastinal, hilar, scalene (ipsilateral or contralateral), supraclavicular (unless ipsilateral superior sulcus)
Distant metastasis (M)	
M0:	No distant metastasis
M1a:	Pleural effusion, pericardial effusion; synchronous nodules in contralateral lung
M1b:	Distant metastasis

Adapted from Rice TW *et al.* Cancer of the esophagus and esophagogastric junction: data-driven staging for the seventh edition of the American Joint Committee on Cancer/ International Union Against Cancer Cancer Staging Manuals. *Cancer* 2010; **116**(16):3763–73.

Table 14.2 Stage grouping for lung cancer

Stage	T	N	M
IA	1a	0	0
	1b	0	0
IB	1a	0	0
IIA	1a	1	0
	1b	1	0
	2a	1	0
	2b	0	0
IIB	2b	1	0
	3	0	0
IIIA	1–2	2	0
	3	1–2	0
	4	0–1	0
IIIB	1–4	3	0
	4	2	0
IV	Any	Any	1

Table 14.3 Regional lymph nodes

Superior mediastinal nodes (N2/3)	
1	Highest Mediastinal
2	Upper paratracheal
3	Prevascular and retrotracheal
4	Lower paratracheal
Aortopulmonary window nodes	
5	Subaortic
6	Para-aortic
Inferior mediastinal nodes (N2/3)	
7	Subcarinal
8	Paraesophageal
9	Pulmonary ligament (Belsey's)
N1 nodes (N3 if contralateral)	
10	Hilar
11	Interlobar
12	Lobar
13	Segmental

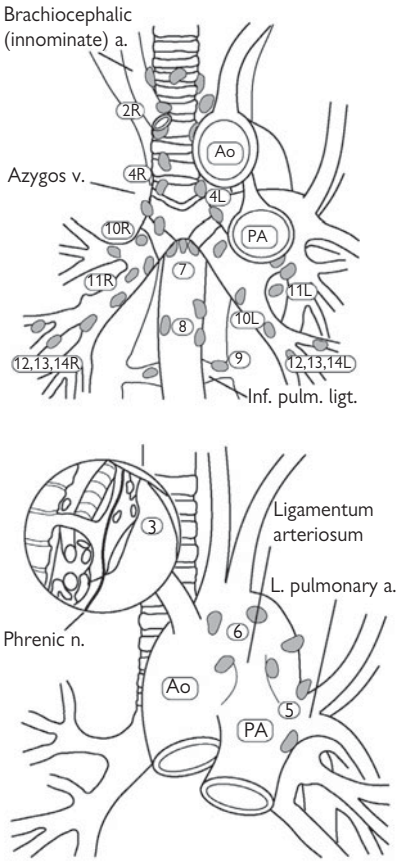


Fig. 14.1 Regional lymph nodes.

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Chemoradiotherapy

Key facts

- For operable patients with stage I and II lung cancer, upfront therapy is surgical resection.
- Platinum-based adjuvant chemotherapy is reserved for pathologic node-positive stage IIA and IIB disease.

Modalities

Neoadjuvant (multimodality, chemoradiotherapy followed by surgery)

- Optimal for stage IIIA, N2 positive lung cancer.
- Superior to surgery alone.
- Multimodality therapy most beneficial for patients with single N2 station, and non-bulky (≤3cm) mediastinal lymph nodes.
- If residual N2 disease, resection is warranted if R0 resection is achievable.[1]
- High mortality if pneumonectomy is required after neoadjuvant chemoradiation:[2]
  - Regimen included two cycles cisplatin/etoposide/concurrent 45Gy XRT randomized to surgery followed by two additional cycles of cis/etop or XRT to 61Gy followed by two additional cycles cis/etop.
  - Significant increase in progression-free survival for trimodality arm.
  - No difference in overall survival between treatment arms.
  - Subset analysis demonstrated significant 5-year overall survival advantage in the lobectomy group (36% vs. 18%) compared to chemoradiotherapy, but a non-significant disadvantage for the pneumonectomy group (22% vs. 24%) compared to definitive chemoradiotherapy.

Definitive chemoradiation

Appropriate for N3 positive stage IIIB disease, and N2 positive stage IIIA patients who are poor surgical candidates, refuse surgery, would require a pneumonectomy for R0 resection, or who have multistation N2 positive nodes, or bulky (>3cm) N2 positive nodes.

Non-small-cell lung cancer

NSCLC is relatively resistant to chemotherapy. Curative resection allows best chance of long-term survival. Surgery is normally offered to all patients with stage I and stage II disease, along with specific patients with stage III disease. Surgical principles of resection are:

- No spillage of cells from primary tumor during resection.
- Entire tumor must be resected by lobectomy, bilobectomy, pneumonectomy or extended resections along with intrapulmonary lymph nodes. Lesser resections proven to have worse outcome, only considered in high-risk patients or small tumors as part of a clinical trial.
- All accessible mediastinal lymph nodes should be excised or sampled to allow complete staging and plan for any adjuvant therapy.
- Frozen section analysis of resection margins to confirm appropriate surgical resection and complete excision of primary tumor.

Chemotherapy

To date no proven benefit from adjuvant therapy in patients with surgically resectable and potentially curative disease. Interest in neo-adjuvant therapy for locally advanced disease.

Radiotherapy

- *Upfront:* used in patients unfit or unwilling to undergo surgery. Also used in those with bulky stage IV disease. Survival benefit in non-surgical patients when combined with concomitant chemotherapy.
- *Adjuvant:* also used as an adjunct to surgery or as palliation.

Small-cell lung cancer

Small-cell lung cancer has usually metastasized by time of presentation. It is sensitive to chemotherapy. Early recurrence is common.

- The mainstay of treatment is chemotherapy.
- Surgery is indicated for very early (clinical stage I) disease.
- Surgery may be combined with chemotherapy.
- In more advanced disease multiagent chemotherapy combined with radiation therapy is indicated.
- Radiotherapy also plays an important role in palliative treatment of SVC obstruction, hemoptysis, recurrent laryngeal nerve palsy, phrenic nerve palsy, pleural effusion, bony lesions, spinal cord compression, cerebral metastases, localized metastatic lesions.

Complications of chemotherapy:

- Myelosuppression: infections, bleeding, anemia.
- Nausea and vomiting, constipation.
- Alopecia.
- Cardiotoxicity, cardiomyopathy.
- Ototoxicity, nephrotoxicity, peripheral and CNS neuropathy.
- Electrolyte disturbances.
- Mucositis.
- Pulmonary fibrosis.
- Second malignancies (acute leukemia, second lung primary).

Complications of radiotherapy

- ***Thoracic:*** esophagitis, pneumonitis, pulmonary fibrosis, myelitis, esophageal stricture, cardiac toxicity.
- ***Cranial:*** tremor, confusion, slurred speech, ataxia, memory deficits, problems with concentration, otitis, scalp erythema, dementia, dysarthria.
- ***Postsurgical:*** thoracotomy, incision pain, wound breakdown, bronchopleural fistulas, respiratory failure.

References

1. Martin J *et al.* Long-term results of combined modality therapy in resectable non-small-cell lung cancer. *JCO* 2002; **20**; 1989-05.
2. Albain *et al.* IASLC 2003. Intergroup 0139 Definitive chemoradiotherapy vs. neoadjuvant for stage IIIA. *Lung Cancer* 2003; **41**(Suppl 2):A#PL-4.

Pulmonary resection overview

Key facts

- Anatomic lobectomy is the standard of care for pulmonary resection of primary NSCLC.
- Lobectomy can be performed through a ***posterolateral thoracotomy***, muscle sparing thoracotomy, axillary thoracotomy, or by ***VATS***.
- Sublobar resection (***segmentectomy*** or non-anatomic wedge) is used for patients with impaired pulmonary function, or small (<2cm) ground glass opacities such as bronchoalveolar carcinoma, or secondary or metastatic tumors to the lung as part of a metastectomy.
- Bronchoplasty or pulmonary arterioplasty (sleeve) procedures should be considered to avoid ***pneumonectomy***.
- A mediastinal lymph node sampling or dissection should be performed as part of any pulmonary resection for primary lung cancer.
- ***VATS lobectomy*** is becoming more prevalent for the resection of early stage lung cancer, less so for larger and central tumors.

General principles

- Patients undergoing lung resection need additional detailed work-up (📖 p650) in addition to the extensive investigations already performed for diagnosis and staging (📖 p672).
- Lung function testing should be performed in all patients, with calculation of ***predicted postoperative respiratory capacity*** (📖 p652), so that resection can be modified (e.g., subloar resection rather than lobectomy) if necessary to reduce the risk of postoperative respiratory insufficiency. Broad rules of thumb are:
  - The minimum acceptable FEV_1 after surgery is 800mL.
  - Pneumonectomy will reduce FEV_1 by around 50%, lobectomy by around 20% depending on the lobe.
  - The preoperative FEV_1 for lobectomy should therefore be >1L, and at least 2L for pneumonectomy.

Anesthesia and surgical set-up

- General ET anesthesia with invasive arterial monitoring.
- A flexible bronchoscopy (📖 p654) is performed via a single lumen tube, before lung isolation is obtained using a bronchial blocker or a double lumen tube.
- The patient is then positioned in the ***lateral, decubitus position*** (📖 p39) with spine parallel to and close to the edge of the table, using sandbags, or rolls and tape, usually with the upper arm positioned on a rest, the lower arm flexed in-front of the head, and the table 'broken' (a little head-down and a little feet-down) so that the hips do not obstruct VATs instruments, and the rib spaces open up a little.
- For ***posterolateral thoracotomy*** (📖 p39) prep and drape so that the spinous processes, ipsilateral nipple, and midaxillary line are exposed.

Mediastinal lymph node dissection and sampling

Accurate tumor staging is critical for prognosis, planning therapy, and clinical studies. Current practice in intraoperative lymph node examination ranges from simple sampling to radical mediastinal lymph node dissection (MLND), with no consensus as to best practice: lymph nodes were sampled in <50% of patients with lung cancer in the USA in one survey. The main debate is between MLND and systematic sampling.

Mediastinal lymph node dissection

- Involves complete removal of most mediastinal lymph nodes for right- and left-sided pulmonary resections.
- *Right-sided resections:* systematic removal of stations 2R, 4R, 7, 8R, and 9R for right-sided resections, with mobilization of SVC.
- *Left-sided resections:* systematic removal of stations 2L, 4L, 5, 6, 7, 8L, and 9L with mobilization of the aortic arch and left subclavian artery.
- May be higher risk of morbidity, e.g., chylothorax, phrenic or recurrent laryngeal nerve injury (beware cautery in the aortopulmonary window), increased mediastinal drainage from bleeding or lymphatics.
- No evidence that MLND contributes to tumor control or survival.

Mediastinal lymph node sampling

- Involves removal of most mediastinal lymph nodes for right- and left-sided pulmonary resections, without the goal of complete removal. All lymph nodes readily visible are removed.
- *Right-sided resections*: involves a systematic sampling of stations 2R, 4R, 7, 8R, and 9R lymph nodes without mobilization of the SVC.
- *Left-sided resections:* systematic sampling of stations 5, 6, 7, 8L, and 9L without mobilization of the aortic arch and left subclavian artery.
- Possibility of less accurate staging, with N2 disease unrecognized in up to 5% of cases.

Evidence base: ACOSOG Z0030

- Randomized 1000 patients to lymph node sampling or lymph node dissection after pulmonary resection.
- No difference in perioperative morbidity or mortality[1] nor long-term survival.

Other technical considerations

- The vessels can be divided in any order:
  - The bronchus of the diseased segment is the landmark used to guide resection: this can be most easily visualized after division of the segmental artery which usually runs alongside it.
  - Dividing the pulmonary artery before the pulmonary veins may help avoid venous engorgement of the resected section, making dissection more difficult, and reduce hematogenous tumor spread.
- Use of clamps and sutures or ligatures or staplers for vessels and bronchus is a matter of surgeon preference.
- Prior to closure the chest is filled with saline, the lung inflated and checked carefully for air leaks, which must be carefully oversewn.

Reference

1. Allen MS *et al.* Morbidity and mortality of major pulmonary resections in patients with early-stage lung cancer: initial results of the randomized, prospective ACOSOG Z0030 trial. *Ann Thorac Surg* 2006; **81**(3):1013–19.

Right lobectomy

Set-up

- The patient is positioned in the left lateral decubitus position, right-side up, with the right lung fully deflated (📖 p39).
- Right posterolateral thoracotomy incision (4th ICS) (📖 p39) or VATS approach (📖 p686).
- Postoperative instructions and management are described on 📖 p696.

Right upper lobectomy

Pulmonary vein division

- The lung is positioned posteriorly, the anterior hilar pleural is incised.
- The right upper lobe (RUL) pulmonary vein is identified. Care is taken to preserve the right middle lobe vein, as it takes off from the inferior border of the proximal superior pulmonary vein.
- The vein may be divided using a vascular stapler, or silk ligature.

Pulmonary artery division

- The right main pulmonary artery (PA) exits posterior to the SVC. The first branch of the right main PA is the truncus anterior, which exits posterior to the SVC.
- The truncus branch is divided with a stapler or silk ligature.
- The final arterial branch of the RUL is the posterior ascending artery. It is exposed through one of three approaches.
  - ***Anterior:*** the upper lobe is retracted posteriorly, and after division of the right upper lobe pulmonary vein and truncus anterior, the right main PA is dissected distally as it becomes the interlobar PA, until the posterior ascending artery is identified lateral and adjacent to the RUL bronchus.
  - ***Central:*** the confluence of the major and minor fissures is dissected exposing the interlobar PA. The proximal take-off of the posterior ascending artery is then clearly identified.
  - ***Posterior:*** the RUL is retracted anteriorly, identifying the take-off of the RUL bronchus from the right main stem bronchus. The station 10R or 'sump' lymph node is removed, revealing the posterior ascending artery.

Bronchus division and completion of fissure

- After division of the vessels, the RUL is retracted anteriorly, and the RUL bronchus is identified as it takes off from the right main stem.
- Peribronchial lymph nodes are removed, and the bronchus is divided with a stapler or with a knife and closed with Vicryl sutures.
- The major and minor fissures are completed with a stapler.

Middle lobectomy

Pulmonary vein dissection

- The lung is positioned posteriorly, the anterior hilar pleural is incised.
- The middle lobe pulmonary vein (MPV) is identified as it takes off from inferior border of the SPV.
- The MPV is divided with a stapler or silk ligature.

Pulmonary artery dissection

- The middle lobe pulmonary artery (MPA) is a branch of the ILPA that is distal and opposite to the PA branch of the superior segment of the lower lobe.
- The MPA can be identified centrally by dissecting the confluence of the pulmonary fissures, or anteriorly after division of the middle lobe bronchus.

Bronchus dissection and completion of fissure

- *Central approach:* the middle lobe bronchus (MB) is identified after division of the anteriorly located MPA.
- *Anterior approach:* the MB is identified first after division of the MPV.
- The MB is divided with a stapler or cut and closed with absorbable suture. Care is taken not to impinge the lower lobe bronchus.
- The inferior major and minor fissures are completed with a stapler.

Right lower lobectomy

Pulmonary vein dissection

- The lung is retracted superiorly, exposing the inferior pulmonary ligament. The ligament is divided with cautery. The station 9 (Belsey's) lymph node will be encountered, and then the inferior pulmonary vein (IPV).
- The IPV is easily encircled and divided with a vascular stapler or silk ligation.

Pulmonary artery dissection

- The ILPA is identified after dissecting the confluence of the fissures.
- The superior segmental artery is encountered extending superiorposteriorly. This is divided.
- The MPA is preserved as it branches off of the ILPA opposite to the superior segmental artery.
- Past the MPA, the ILPA becomes the common basal artery, which is divided by a vascular stapler or silk ligature.

Bronchus dissection

- After division of the common basal artery, the bronchus is identified posterior to this vessel, and divided.
- Care is taken not to impinge the MLB.
- Completion of the fissure.
- The major and minor fissures are completed with a stapler.

Left lobectomy

Key facts

- The patient is positioned in the right lateral decubitus position, left side up, with the left lung fully deflated (📖 p39).
- Right posterolateral thoracotomy incision (4th ICS) (📖 p39) or VATs approach (📖 p696).
- Postoperative instructions and management are described on 📖 p696.

Left upper lobectomy

Left upper lobectomy is the most technically demanding of any of the lung resections because of the location of the PA in relation to the aortic arch, and the PA branches—particularly the ***anterior segmental artery***.

Pulmonary vein dissection

- The lung is retracted posteriorly, and the pleura anterior to the hilum is incised, exposing the superior pulmonary vein (SPV), as it lies anterioinferior to the left main PA.
- The SPV is encircled and divided with a vascular stapler or silk ligation.

Pulmonary artery dissection

- The left main PA enters superiorposterior to the SPV, and travels superioranterior to the left main bronchus, and passes posterior to the LUL bronchus as it becomes the interlobar PA.
- There may be multiple segmental branches extending from the interlobar PA to the LUL.
- Exposure of the interlobar PA and segmental branches is performed both with dissection anteriorly after division of the SPV, and by exposing the interlobar PA by dissecting the interlobar fissure.
- The anterior segmental artery and any accessory anterior branches are divided anteriorly: the anterior segmental artery is known as the '***artery of sorrow***'. It is a short but wide branch of the left main PA that can be avulsed by over-vigorous retraction, and if the tear extends proximally may force a pneumonectomy (see 📖 p690).
- The apical, posterior, lingular and other accessory arterial branches are divided after exposure of the ILPA. Care is given to preserve the posterior segmental artery of the lower lobe, branching posterior and opposite to the take-off of the lingular PA segments.

Completion of the fissure and bronchus dissection

- The superior and inferior interlobar fissure is completed with a stapler.
- After division of the vessels and fissure, the LUL bronchus is clearly identified and divided with a stapler.

Pulmonary arterioplasty

- Tumors involving the inner curvature of the interlobar PA may require a PA arterioplasty.
- The proximal and distal control is obtained of the left main PA and interlobar PA. If the distal interlobar PA cannot be controlled, the SPV is divided, and the IPV is clamped, and this should markedly reduce back bleeding.

- The tumor is sharply excised, and the PA defect is either closed primarily, or using an autologous or bovine pericardial patch.

Left lower lobectomy

Pulmonary vein dissection

- The lung is retracted superiorly, exposing the inferior pulmonary ligament. The ligament is divided with cautery. The station 9 (Belsey's) lymph node will be encountered, and then the inferior pulmonary vein.
- The inferior pulmonary vein is easily encircled and divided with a vascular stapler or silk ligation.

Pulmonary artery dissection

- The ILPA is identified after dissecting interlobar fissure.
- The superior segmental artery is encountered extending superiorposteriorly, and opposite to the lingular branches. This is divided.
- Past the lingular branches, the ILPA becomes the common basal artery, which is divided by a vascular stapler or silk ligature.

Bronchus dissection and completion of the fissure

After division of the common basal artery, the bronchus is identified posteriorly, and divided, and the fissure is completed with a stapler.

Injury to the PA during left upper lobectomy

This is a dreaded complication due to ***rupture of the anterior segmental artery from the left main PA*** which may lead to unplanned pneumonectomy, and rarely to the on-table death of a patient from uncontrollable hemorrhage. It can result from traction injuries, or directly from cautery, or loss of control of ligatures after branch vessel division.

- If circumferential control of the vessel was not obtained prior to injury, trying to get a clamp around a hemorrhaging left main PA will result in torrential blood loss and is not advisable.
- Attempting to get a clamp over the laceration is difficult without circumferential control of the vessel, which may tear further, and will result in torrential blood loss.
- Blindly suturing a bleeding left main PA is virtually impossible, usually resulting in further damage to the vessel and torrential blood loss.

Management

- Put a finger on the hole, gently. If the hole is much bigger than your finger, or too deep, put a folded-up sponge on a sponge holder on the hole. This will buy you an indefinite amount of time.
- Tell your anesthesiologist that you have a serious problem: do not proceed further until you have blood in the room.
- Get senior help. Once you have help extend the incision, while maintaining control of the PA, so that you have adequate access.
- Now open the pericardium and gain circumferential control of the intrapericardial portion of the left superior pulmonary artery: clamping this will allow you to divide and oversew the avulsed branch.
- Always be wary of a hematoma on this branch artery which may be the only sign of a traction injury, heralding rupture: get proximal control and divide it as described in this chapter.

VATS lobectomy

Key facts

- A spectrum of procedures are termed VATS (video-assisted thoracoscopic surgery) ranging from totally endoscopic approaches without a mechanical retractor or rib spreading, to those performed though a minithoracotomy, assisted by a camera.
- Some surgeons advocate simultaneous stapling of all the hilar structures, rather than anatomical lobectomy: this is not described here.
- Many centres routinely perform complete anatomic lobectomy using a thoracoscopic approach, with equivalent oncological outcomes to the open technique, and reduced surgical trauma, postoperative pain, length of stay, chest tube duration, preserved pulmonary function, and improved cosmesis.

Patient selection

- The indications for VATS lobectomy are the same as for the open procedure, i.e., known or suspected lung cancer amenable to lobectomy.
- Similar cardiopulmonary capacity is required (📖 p652), although it has been suggested that older, frailer patients who are not candidates for an open lobectomy may benefit from a VATs approach.
- ***Contraindications:*** inability to provide complete resection with a lobectomy, T3 (chest wall), and T4 tumors, N2 and N3 disease, inability to achieve or tolerate single-lung ventilation.
- If the tumor is visualized in the lobar orifice at bronchoscopy suggesting need for a sleeve resection, then most surgeons would opt for an open procedure.
- Surgeon experience dictates whether patients with prior thoracic surgery or adhesions, T3 tumors involving the pericardium or diaphragm, incomplete fissures are suitable for a VATS approach.

Procedure

Set-up

- General endotracheal anesthesia with invasive arterial monitoring.
- A flexible bronchoscopy (📖 p654) is performed via a single lumen tube, before lung isolation is obtained using a bronchial blocker or a double lumen tube,
- The patient is then positioned in the *lateral, decubitus position* (📖 p39) with spine parallel to and close to the edge of the table, using sandbags, or rolls and tape, usually with the upper arm positioned on a rest, the lower arm flexed in-front of the head, and the table 'broken' (a little head down and a little feet down) so that the hips do not obstruct VATS instruments, and the rib spaces open up a little.

Port placement

- This is very variable but generally follows similar basic principles:
- The first port is a 5–10mm port (depending on the size of camera used) for the camera, placed in the midaxillary line, in the 7th or 8th intercostal space, which gives good visualization of the hilum and a

panoramic view of the hemithorax. Always align the camera, pathology, and monitor to reduce loss of automatic hand–eye coordination.
- The second incision is a working port—usually a 3–5cm incision in the 5th or 6th intercostal space in the inframammary crease. This can be opened with a soft tissue retractor, or potentially a small rib-spreader.
- Additional 5mm incisions for instruments are sometimes placed posteriorly or in the axilla, aiming to triangulate the ports, but it is possible to complete a VATS lobectomy through those two incisions.

Equipment
- A 30° or 45° angled or flexible scope to provide a panoramic view of the operative field without competing with the other instruments.
- Conventional sponge holders, forceps, and scissors can often be used through the working port.
- Endoscopic instruments include forceps, scissors (with the option of bipolar cautery), and soft-tipped (peanut) instruments for blunt dissection, as well as endoscopic suction tips and specially designed endoscopic stapling device, e.g., EndoGIA vascular (2–2.6mm) and bronchial (3.5mm) staplers.
- The specimen pouch is a bag in which the specimen can be placed while inside the chest, facilitating removal of a bulky tissue through a small incision without seeding malignant cells into the wound.

Dissection
- Carefully confirmation of the tumor location and lobe: manual palpation is usually not possible poor alignment of camera, pathology, and monitor may disorient the surgeon, so this step is done with care.
- The pulmonary vein is mobilized using gentle blunt dissection, with the lung retracted posteriorly (and inferiorly for upper lobectomy, superiorly for lower lobectomy) under direct visualization through the working port.
- The pulmonary vein is stapled and divided, exposing the pulmonary artery which is mobilized, stapled, and divided.
- The bronchus is usually then stapled and divided and the fissure completed with staples.
- The specimen is removed using a specially designed bag, and the hemithorax irrigated with water, and the bronchial stump and fissures inspected for air leaks.

Closure
- Chest tubes are placed (one apical, one basal) via the instrument port sites, together with a costal block.
- Soft tissues are closed in layers with interrupted absorbable sutures.

Postoperative care
- Drains on $-20cmH_2O$ waterseal suction, and CXR in recovery area.
- This is much the same as for open lobectomy, except epidural and other analgesic requirements are less, and patients should be able to mobilize and be discharged earlier.

Sleeve resection

Key facts

- Patients with tumor extending out from the upper lobe orifice may require a ***sleeve resection*** to avoid pneumonectomy.
- The patient is positioned in the lateral decubitus position, ipsilateral side up with the lung fully deflated (📖 p39).
- Posterolateral thoracotomy incision (4th ICS) (📖 p39).
- Postoperative instructions and management are described on 📖 p696.

Right upper lobectomy sleeve

- After division of the RUL arteries and veins as described on 📖 p682, the right main bronchus is divided with a knife proximal to the tumor, and the bronchus intermedius divided distal to the tumor (slightly obliquely to minimize size discrepancy between the right main bronchus and the bronchus intermedius orifices).
- Specimen is sent for frozen section pathology to confirm R0 resection.
- 2/0 Vicryl sutures are placed in the right main bronchus and bronchus intermedius.
- The two ends are reanastomosed with either interrupted or running absorbable suture (Vicryl or PDS): a hilar release may be needed to ensure a tension free reanastomosis.
- Tumors at the base of the truncus anterior may require PA arterioplasty.
- The proximal and distal control is obtained for the right main PA and interlobar PA.
- The tumor is sharply excised.

Bilobectomy sleeve

- Patients with tumor occluding the middle lobe orifice or bronchus intermedius, may require a middle and right lower lobe bilobectomy.
- After ligation of the middle and right lower lobe pulmonary vessels (📖 p682) and completion of the fissures, the lung is retracted anteriorly, exposing the right main, right upper bronchus, and bronchus intermedius.
- The right main bronchus is divided with a knife, as well as the RUL bronchus, and the specimen is removed.
- The RUL bronchus is cut slightly obliquely to minimize size discrepancy between the right main bronchus and the RUL orifices.
- 2/0 Vicryl sutures are placed in the right main bronchus and the bronchus intermedius.
- The two ends are reanastomosed with either interrupted or running absorbable suture (Vicryl or PDS).

Left upper lobectomy sleeve

- After the division of the left upper lobe arteries and veins, the interlobar PA is encircled with umbilical tape and retracted laterally.
- The left main bronchus is divided with a knife proximal to the tumor, and the left lower lobe bronchus is divided distal to the tumor.
- The specimen is sent for frozen section pathology to confirm R0 resection.
- The left lower lobe bronchus may be cut slightly obliquely to minimize size discrepancy between the right main bronchus and the bronchus intermedius orifices.
- 2/0 Vicryl sutures are placed in the right main bronchus and bronchus intermedius.
- The two ends are reanastomosed with either interrupted or running absorbable suture (Vicryl or PDS).
- A hilar release may be performed to ensure a tension-free reanastomosis, which may be covered with an intercostal muscle pedicled flap.

Left lower lobe sleeve lobectomy

- Patients with tumor occluding the lower lobe orifice approaching the left lower lobe bronchus may require a sleeve resection to preserve the left upper lobe, and avoid a pneumonectomy.
- After ligation of left lower lobe pulmonary vessels and completion of the fissures, the lung is retracted posteriorly, exposing the left main bronchus, left upper, and left lower lobe bronchus.
- The left main bronchus is divided with a knife, as well as the left upper lobe bronchus, and the specimen is removed.
- The left upper lobe bronchus may be cut slightly obliquely to minimize size discrepancy between the left main bronchus and the left upper lobe orifices.
- 2/0 Vicryl traction sutures are placed in the left main bronchus and left upper lobe bronchus.
- The two ends are reanastomosed with either interrupted or running absorbable suture (Vicryl or PDS).

Pneumonectomy

Key facts

- Right pneumonectomy carries the highest mortality, and morbidity, especially bronchopleural fistula (BPF).
- The bronchial stump should always be covered with tissue, including intercostal muscle flap, pericardial fat pad, or pleural flap.
- Left pneumonectomy carries lower mortality and morbidity compared to right, with less incidence of BPF because, although the left main bronchus is longer than the right main bronchus, the stump retracts into mediastinal tissue where it is covered by the arch of the aorta.

Right pneumonectomy

Pulmonary vein dissection

- The lung is positioned posteriorly and the anterior hilar pleura is incised.
- The inferior pulmonary ligament is divided with cautery up to the right inferior pulmonary vein.
- The right superior and inferior pulmonary veins are divided using a vascular stapler, which can be done safely without subsequent engorgement of the lung.
- If tumor approaches the veins, making division difficult, the pericardium may be opened, gaining direct access to the pulmonary veins at the exit of the left atrium. The veins can be divided at this point, or if necessary the left atrium wall may be clamped using a vascular clamp, the confluence of the veins divided, and the atriotomy closed with a running Prolene suture in two layers.

Pulmonary artery dissection

- RMPA exits posterior to the SVC, and the SVC is mobilized and retracted medially to expose the RMPA. The RMPA is carefully dissected and encircled, and doubly ligated with vascular stapler.
- If tumor is to close to the RMPA as it exists posterior to the SVC, the SVC can be mobilized laterally, and the proximal RMPA can be exposed as it exits posterior to the aortic arch, by dividing the pericardium anterior to the phrenic nerve.

Bronchus dissection

- The RMB is then stapled or divided close to the carina, to avoid a long RMB stump, and subsequent pooling of secretions.
- The bronchial stump should be covered with tissue as mentioned earlier.

Completion of the operation

- All pericardial defects should be closed primarily or with bovine pericardium or prosthetic material to prevent cardiac herniation.
- Careful hemostasis is obtained.
- A chest tube is not needed.
- Careful watertight wound closure.

Left pneumonectomy

Pulmonary vein dissection

- The SPV and IPV are divided in similar fashion as in a right pneumonectomy.

Pulmonary artery dissection

- The LMPA exits posterior to the arch of the aorta.
- The lung is retracted posteriorly, and the pleura overlying the mediastinum is divided below the aortic arch, anterior to the vagus nerve and recurrent laryngeal nerve.
- The LMPA is carefully dissected, encircled and doubly ligated.
- If the tumor extends proximally, control of the LMPA can be obtained dividing the ligamentum arteriosum, mobilizing the aortic arch superior and laterally, and incising the pericardium and controlling the intrapericardial origin of the LMPA.

Bronchus dissection

- To gain access to the LMB, the aortic arch is mobilized and retracted superiorly and laterally.
- The LMB is then stapled or divided close to the carina as possible by pushing the stapler against the aortic arch.
- The bronchial stump should be covered with tissue as mentioned previously in this chapter (📖 p690).
- Completion of the operation.
- All pericardial defects should be closed primarily or with bovine pericardium or prosthetic material to prevent cardiac herniation.
- Careful hemostasis is obtained.
- A chest tube is not needed.
- Careful watertight wound closure.

Superior sulcus tumors

Key facts

- Superior sulcus tumors are essentially apical lung tumors that invade the chest wall, and so by definition these are all T3 tumors.
- ***Pancoast tumors*** are superior sulcus tumors, but a superior sulcus tumor is not necessarily a Pancoast tumor.
- Curative resection involves adjuvant chemoradiotherapy and en bloc resection including chest wall resection, possible hemi-vertebrectomy, and possible excision of the T1 nerve root of the brachial plexus.

Presentation and investigation

- Presentation depends on the location of the tumor: the thoracic inlet can be divided into three compartments by the insertion of scalenus on the 1st and 2nd ribs:
  - Anterior compartment contains platysma, SCM, and omohyoid muscles; jugular and subclavian veins: symptoms include chest wall pain, hand or arm swelling from subclavian vein compression.
  - Middle compartment includes anterior and middle scalene muscle, phrenic nerve, subclavian artery and branches, brachial plexus trunks: symptoms include brachial plexus pain and paresis.
  - Posterior compartment includes spinal accessory nerves, sympathetic chain and stellate ganglion, vertebral bodies and intercostal nerves: symptoms are those of Pancoast tumors.
- This is as described for lung cancer (📖 p672). The apex of the lung is difficult to image on plain CXR, and patients whose primary complaint is shoulder pain are often misdiagnosed until CT chest is performed.
- Defining vascular and neurological invasion is essential for planning and resection and reconstruction: clinical evidence of T1 and C8 involvement suggests R0 resection may leave patient with severe ulnar paresis.
- Contraindications to resection are invasion beyond C8 and T1; multiple vertebral involvement with spinal canal invasion, N2 and N3 disease.

Pancoast tumors

Pancoast tumors are posteriorly located, and lead to ***Pancoast syndrome*** due to invasion into the ***stellate ganglion*** and ***brachial plexus***:

- They are characterized by pain in shoulder extending to the ulnar aspect of the forearm and hand.
- Atrophy of the hand and arm muscles.
- Horner's syndrome (ptosis, miosis, and anhidrosis).
- Tumor spread occurs along nerve roots up to spinal canal.

Management of superior sulcus tumors

- Ipsilateral supraclavicular lymphadenopathy is considered N1 disease.
- Operative candidates should undergo the SWOG protocol: induction chemoradiotherapy (etoposide, cisplatin, 45Gy).
  - Patients with stable disease or disease regression then undergo anatomical resection, with two more cycles of adjuvant chemotherapy.
  - Around 90% of T3 and T4 tumors have R0 resection.
  - 4% operative mortality, 55% of patients had a complete pathologic response, overall 5-year survival was 41%, 53% for R0 resection.

Surgical approaches

Set-up

- General ET anesthesia with invasive arterial monitoring, and usually a wide-bore central line allowing rapid large volume infusion.
- A flexible bronchoscopy (📖 p654) is performed via a single lumen tube, before lung isolation is obtained.
- The patient is positioned and lung isolation rechecked.

Posterior approach (Shaw–Paulson)

- For posterior lesions: position patient in lateral decubitus (Fig. 14.2) with base of skull, C7 to iliac crest prepped and in field.
- Initial posterior-lateral thoracotomy (📖 p39) to ascertain resectability.
- Can be combined with chest wall resection and hemi-vertebrectomy.
- Incision is extended between lateral border scapula and spinous processes to C7 dividing trapezius along its full length, rhomboids (preserving dorsal scapular nerve). Retractor placed between interspace and tip of scapula, and subscapular muscles divided up to 1st rib.
- En bloc chest wall resection: posteriorly erector spinae incised along its length to expose angle of ribs and transverse processes which can be resected or disarticulated depending on tumor invasion.
- Bleeding from vertebral foramen is dealt with by loose packing.
- The brachial plexus and subclavian vessels are dissected: the subclavian artery may be reconstructed, whereas the vein is ligated if involved.
- Vertebrae are resected and reconstructed with methylmethacrylate and plate-and-screw locking systems; and anatomic upper lobectomy performed with the chest wall specimen still attached.

Anterior approach (Dartevelle or hemiclamshell)

- Used for anteriorly located tumors: can be combined with an anterior thoracotomy, extended into the neck anterior to the sternocleidomastoid muscle, and laterally into the deltopectoral groove.
- Sternocleidomastoid is reflected anteriorly to expose the entire neck and thoracic inlet, except structures behind medial clavicle which may be resected to improve exposure: resecting the sternoclavicular joint is avoided to preserve shoulder movement postoperatively.
- After chest wall resection an anatomic lobectomy is performed (📖 p682).
- *Chest wall reconstruction* is described on 📖 p808.

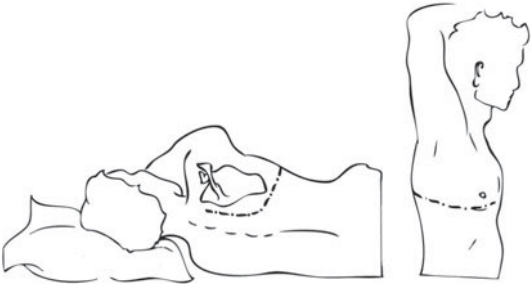


Fig. 14.2 Shaw–Paulson approach to superior sulcus tumors.

Sublobar resection

Key facts

- Sublobar resection (non-anatomic ***wedge resection*** or anatomic ***segmentectomy***) preserve lung parenchyma in patients with small and peripheral tumors who require surgery for control, but are judged to have insufficient pulmonary reserve to tolerate lobectomy.
- Other indications include localized lesions, e.g., bronchiectasis, aspergillosis, TB, and metastatic lung lesions.
- Non-anatomical resection is an oncological compromise: around 1/3 of patients will experience local recurrence, but with careful patient selection the negative impact on oncologic outcomes may be minimal.

Principles of segmentectomy

- A wedge resection is a non-anatomic resection with a stapler, achieving wide margins. Anatomic segmentectomy is based on the segmental anatomy of the bronchial tree (Fig. 13.2, 📖 p655).
- Generally lesions suitable for sublobar resections should be in the outer 1/3 of the lung parenchyma and <3cm in size.
- Segmentectomy is technically more demanding than wedge resection, but may be preferable for primary lung tumors because:
  - More reliable deep margins as resection extends to hilum.
  - Lymphatic drainage and interlobar nodes included.
- Wedge resection are technically easiest when the lesion is near an acute lung margin, otherwise tissue becomes too thick to staple.

Procedure

Left upper lobe apical trisegmentectomy

- Involves resection of the anterior (S3) and apicoposterior (S1+2) segments (upper division).
- The upper division vein is mobilized as it branches from the left superior pulmonary vein, and divided, the lingular vein is preserved.
- The segmental pulmonary arteries branching to the upper division are identified anteriorly, posteriorly, and within the fissure, and divided.
- The upper division bronchus is identified within the fissure, with preservation of the lingular bronchus.
- The lung is partial inflated to identify the demarcation of the segmental fissures for division.

Lingulectomy

- Resection of the superior (S4) and inferior (S5) lingular segments.
- The lingular vein is mobilized as it branches from the left superior pulmonary vein, and divided. The upper division vein is preserved.
- The fissure between the upper and lower lobe is completed.
- The lingular artery is identified and divided.
- The lingular bronchus is dissected and divided, with preservation of the upper division bronchus.
- Any additional lingular arteries are identified and divided.
- The lung is partially inflated to identify the demarcation of the remaining segmental fissure for division.

Right or left superior segmentectomy

- Involves resection of the superior segment of the lower lobe (S6).
- The major fissure is completed.
- The superior segmental artery is found within the fissure and divided.
- Posterior to the artery, the superior segmental bronchus is dissected and divided.
- The superior segment vein is identified as it branches off of the inferior pulmonary vein and divided.
- The remaining segmental fissure is identified with partial inflation of the lung, and divided.

Right upper lobe anterior segmentectomy

- Involves resection of the anterior segment (S1).
- The minor fissure is completed.
- The inferior edge of the superior pulmonary vein is dissected. The anterior segmental vein branches before the take-off of the middle lobe vein. The anterior segmental vein is then divided.
- The anterior segmental artery is then identified and divided.
- The anterior segmental bronchus is posterior to the artery. Once dissected, it is divided.
- The remaining segmental fissures are identified with partial inflation of the lung, and divided.

Oncologic outcomes for sublobar resection

- Lung Cancer Study Group randomized sublobar resection versus lobectomy for stage IA lung cancer and found a 17-fold increase in local recurrence in the sublobar cohort (17% vs. 6%).[1]
- However, tumors that are peripheral and <2cm in size appear to have oncological equivalence for sublobar resection versus lobectomy.[2]
- Segmentectomy versus lobectomy: no difference in 5-year survival for tumors <2cm, but survival advantage with lobectomy for tumors >2cm.[3]
- Outside of a clinical trial, lobectomy is the standard of care in the medically operable patient with early stage lung cancer.

References

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2. Okada M *et al.* Effect of tumor size on prognosis in patients with non-small cell lung cancer: the role of segmentectomy as a type of lesser resection *J Thorac Cardiovasc Surg* 2005; **129**:87–93.
3. Fernando HC *et al.* Lobar and sublobar resection with and without brachytherapy for small stage IA non-small cell lung cancer. *J Thorac Cardiovasc Surg* 2005; **129**:261–7.

Care after lung resection

Key facts

- Mortality after pulmonary resection is <2% for lobectomy and 6% for pneumonectomy.
- Morbidity for pulmonary resection is ~25% for both.
  - Perioperative morbidity is primarily cardiovascular and respiratory: pneumonia, atelectasis, supraventricular tachycardias, myocardial infarction, DVT, and PE.
  - Long-term complications include bronchopleural fistula, prolonged air leak, post-thoracotomy pain syndrome, and oxygen dependence.

Preoperative strategies to reduce morbidity

- Preoperative physiotherapy, including use of an incentive spirometer.
- Maximizing preoperative nutrition, especially for patients with preoperative cachexia and anorexia, and who have lost >10% of body weight (which is associated with increased mortality).
- Optimization of preoperative comorbidities, including glucose control for diabetics and bronchodilator therapy for patients with COPD, and smoking cessation.
- Instructing patients to walk 1–2 miles a day, to increase lower extremity circulation, and lung capacity.

Postoperative management

Monitoring

- Given the high (up to 15%) incidence of cardiac arrhythmia, all postoperative patients who do not need ICU care should be on a telemetry floor.
- In high-risk cases invasive pressure monitoring is kept for 48h.
- Patients with epidurals keep their urinary catheter, and may benefit from invasive arterial monitoring for the first 24h.
- Daily CXR: in pneumonectomy fluid level should rise, not fall.

Analgesia

- Epidural or paraspinous (📖 p648) catheters should be used for patients undergoing thoracotomy. Hypotension, bradycardia, and low urine output may be a sign of excessive and high epidural: check sensation of dermatomes which should be intact below the nipple, and reduce or stop epidural if necessary.
- Intermittent opiates are usually sufficient for patients only undergoing VATS.
- NSAIDS such as iv ketorolac or oral ibuprofen are effective in conjunction with narcotics. Iv acetaminophen has a lower side effect profile and similar effectiveness to opiates, and is increasingly used (📖 p648).

Antibiotic prophylaxis

- Reduces incidence of wound infections, but no proven efficacy against pneumonia or empyema.
- IV antibiotic with adequate Gram +ve and some Gram −ve coverage (e.g., cefazolin) at time of induction.
- Does not need to be continued postoperatively.

DVT prophylaxis

- Cancer patients are at high risk for postoperative thromboembolism. Generally 5000U of LMWH should be given subcutaneously prior to induction, and continued postoperatively until the patient is completely ambulatory.
- Sequential alternating leg compression devices (SCDs) can be used in conjunction with prophylactic heparin.

Respiratory physiotherapy

- Pulmonary physiotherapy is critical, including:
- Scheduled use of an incentive spirometer.
- Ambulating 4 times/day.
- Use of a pulmonary physiotherapy standardized protocol is helpful.
- **UC Davis Thoracic Pulmonary Hygiene Pathway.**
- Upon admission to the postanesthesia care unit (PACU), the respiratory therapist will begin positive airway pressure treatment, e.g., EzPAP® with 3% hypertonic saline nebulizer every 4h while in the PACU.
- Upon transfer to the thoracic surgery telemetry floor, the patient will continue with the EzPAP treatments every 6h for the first 48h.
- If the morning CXR shows increasing atelectasis than the EzPAP® will be changed to intrapulmonary percussive ventilation every 6h, which delivers rapid, percussions of oxygen into the lungs while simultaneously delivering therapeutic aerosols to loosen and help propel secretions.
- After 48h, the patient is reassessed using a cardiopulmonary therapy risk factor analysis, and their therapy is reduced as appropriate.

Fluid management

- Excess fluid restriction and administration should both be avoided in patients undergoing pulmonary resection.
- Intraoperative lung manipulation may lead to increased alveolar fluid volumes due to disruption of the alveolar capillary membranes. Excess fluid administration may result in pulmonary edema, atelectasis, and hypoxia.
- Excess fluid administration may also result in prolonged chest tube output.
- A 1500mL fluid restriction is recommended, with early oral intake, and elimination of maintenance IV fluid.

Chest tube management

- Chest tubes drain fluid and potential air leaks.
- Chest tubes are removed generally when output is <240mL/24h and there is no air leak.
- Patients with prolonged air leak (>5 days) may be discharged to home with a Heimlich valve if the lung stays expanded on water seal (see 🕮 pp250–251).
- Excess fluid administration may also result in prolonged chest tube output.
- A 1500mL fluid restriction is recommended, with early oral intake, and elimination of maintenance IV fluid.

Complications after lobectomy

Key facts

These are the commonest complications and can be devastating: prevention and early treatment is absolutely key

Prevention

- Careful preoperative assessment of cardiopulmonary function, predicted postoperative respiratory capacity, and tailoring surgical resection appropriately (📖 pp652–653).
- Optimizing preoperative nutrition and respiratory status.

Reducing perioperative aspiration risk

- Meticulous attention to fasting, gastric transit, and decompression in immediate perioperative period.
- Careful operative technique avoiding direct and traction injuries of the recurrent laryngeal nerves.
- Efficient extubation in the perioperative period, close monitoring of oral diet, and sitting patient up especially when eating or drinking.
- Use of H_2 antagonists.
- Very low threshold for dysphagia diet and assessment, particularly in hoarse, elderly, and debilitated patients, and those with borderline lung function.

Reducing risk of postoperative chest infection

- Mobilize early and aggressively: the patient should not lie flat if they can sit up, not sit in bed if they can sit in a chair, and not sit in a chair if they can stand and walk.
- Good pain control, nursing, and physiotherapy, as well as patient education are key, along with being aggressive about removing lines, catheters, and drains as soon as possible.
- Daily physiotherapy, including incentive spirometry.

Pneumonia

Usually occurs within a week following surgery. Commonly Gram –ve bacilli, although others routinely colonizing the airway of patients with obstructive lung disease such as *Haemophilus influenza* and *Pneumococcus*. Covered in more detail on 📖 p268.

Diagnosis

- Usually productive cough, offensive sputum, temperature, elevated white cell count, and consolidation on CXR.
- If CXR and clinical picture unclear (patient may not have a productive cough or temperature, and the CXR may look more like a collection), CT chest without contrast will demonstrate any consolidation.

Management

- Aggressive physiotherapy, appropriate antibiotics, oxygen therapy if indicated, mobilization, adequate nutrition.
- Early bronchoscopy allows cultures to be taken and more efficient pulmonary toilet.

Respiratory failure

This is a major cause of perioperative mortality. It usually occurs within 4 days of surgery. Patients at high risk should be identified preoperatively. This is covered in more detail on 📖 p160 and 📖 pp162–163.

Management

- Postoperative chest physiotherapy decreases postoperative pulmonary complications and should be part of the normal postoperative management of thoracic surgical patients.
- Retained bronchial secretions require efficient and effective removal, sometimes necessitating the insertion of a minitracheostomy, to prevent atelectasis, hypoxia, and respiratory failure.
- Prolonged management with BiPAP or CPAP is often counterproductive: intubation and ventilation may be required.

Acute lung injury

This is a spectrum of lung injuries that includes ***postpneumonectomy lung syndrome*** and ***adult respiratory distress syndrome*** (ARDS) characterized by respiratory insufficiency and acute onset, bilateral pulmonary infiltrates, hypoxemia. It is covered in more detail on 📖 p163.

- Diagnostic criteria for ARDS: PCWP <18mmHg, non-cardiogenic pulmonary edema, PaO_2/FiO_2 <300 regardless of PEEP.
- Management: fluid restriction and diuresis, low tidal volume, high PEEP.

Arrhythmias

The most common arrhythmia following thoracic surgery is AF (📖 pp263–263). It is more common following pneumonectomy than lobectomy, and is associated with significantly increased mortality. The onset is usually within 48h of surgery. Ventricular arrhythmias are rare. Routine prophylactic administration of antidysrhythmics is not indicated.

Management

- Aim to keep potassium 4–5mmol/L.
- Administration of digoxin is the mainstay of treatment for AF. It may be given orally in patients who are hemodynamically stable or iv in those who have some degree of compromise.
- In more urgent cases administration of a 2.5–5mg iv metoprolol may produce rapid slowing of the ventricular rate.
- If hemodynamically unstable patients may require cardioversion.
- Following this digoxin should be administered to reduce the risk of recurrent AF.
- Almost all patients are back in sinus rhythm 6 weeks following surgery and medication for their dysrhythmia may then be discontinued.

Pulmonary embolism

Reported incidence is low after pulmonary surgery, but mortality is high. Thoracic surgical patients have multiple risk factors: increased age, malignancy, prolonged duration of anesthesia, smoking history, heart disease. They represent a moderate risk group for development of pulmonary embolus. Presentation and management is discussed on 📖 p476.

Outcomes of surgery for NSCLC

Key facts

- Survival after treatment for NSCLC is dependent on treatment modality and pathologic stage (Table 14.4).
- Majority of lung cancers recur within the first 3 years after pulmonary resection, and present as distant metastasis.

Recurrence

- Majority of lung cancers recur within the first 3 years after pulmonary resection, and present as distant metastasis.
- Tumors that recur after 5 years are most likely second primaries.
- The risk of developing a second lung primary tumor after pulmonary resection is 1–2% per year, with the cumulative risk 5% after 5 years and 12.6% after 10 years.

Table 14.4 NSCLC current therapy and 5-year survival[1–3]

Pathologic stage (6th and 7th AJCC)	Therapy	% 5-year survival
I	Surgery	60–75
II	Surgery	30–50
IIIA	Surgery + multimodality therapy	10–40
IIIB	Chemotherapy/radiation	<5–10
IV	Chemotherapy	<1–10

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2. Minna JD. Neoplasms of the lung. In Kasper E (ed) *Harrison's Principles of Internal Medicine*, 16th edn. New York: McGraw-Hill, 2005.

3. Goldstraw P *et al.* The IASLC lung cancer staging project: proposals for the revision of the TNM stage groupings in the forthcoming (seventh) edition of the TNM classification of malignant tumours. *J Thorac Oncol* 2007; **2**(8):706–14.

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Bronchopleural fistula

Key facts

- A ***bronchopleural fistula*** (BPF) is a communication between the bronchial airways and the pleural space, usually after pneumonectomy and in the setting of postoperative empyema.
- The incidence of ***empyema*** following pneumonectomy is 1%, and may develop early (within 3 months following surgery) or many years following surgery.
- The aims of treatment of both empyema and bronchopleural fistula after lung resection is to drain the fluid, and sterilize the space; and in the case of BPF to close the communication with the bronchial airways: surgery is usually required.

Empyema after lung cancer resection

Early empyema may be ***primary*** (bacterial contamination at time of surgery) or ***secondary*** (infection of pleural fluid from infected residual lung tissue or from bronchopleural fistula). Late infection is usually hematogenous in origin. Staphylococcus is the most common infection. Streptococcus and Gram–ve and anerobic organisms also commonly cultured. Management is described in more detail on 📖 p704.

Management

- Early drainage of pleural space using tube thoracostomy, antibiotic therapy, and eventual obliteration of space.
- Bronchoscopy can confirm the presence or absence of bronchopleural fistula.
- A surgical window may be required in order to provide adequate drainage for obliteration of the space.
- Closure of window is possible following complete sterilization of the thoracic cavity.
- Other options include thoracoplasty (📖 p704) or obliteration of the empyema space.

Bronchopleural fistula

- Incidence: 1.5–6%. More common following lung resection for malignancy. Also associated with pre- and postoperative radiotherapy and pre-existing diabetes mellitus.
- More common following pneumonectomy than lesser procedures.
- Right-sided much more common than left-sided.

Prevention

- Limiting the dissection around a bronchial stump limits post-surgical ischemia, and stump breakdown.
- The bronchial stump after pneumonectomy should be covered by viable tissue, e.g., pedicled intercostal muscle, pedicled pericardial fat pad, pleural flap, omentum.
- The left pneumonectomy stump is not always covered, as it is often naturally covered by the arch of the aorta and pulmonary artery.

Diagnosis

- Patient may start expectorating foul watery fluid—most frequently after lying with operated side uppermost. Sometimes patients complain of new 'gurgling' sensation in chest. Signs of pneumonia.
- New or falling air-fluid level in pneumonectomized pleural space on CXR.
- Bronchoscopy confirms diagnosis: allows the stump and any defects in it to be visualized directly, as well as permitting bronchial toilet.

Management

Management first involves preventing the patient from aspiration of pleural fluid through the BPF, and treating the infection

- Nurse the patient with the operated side lowest to prevent spillage into the airway. This is a contraindication to CPAP.
- *Immediate chest tube* after discovery of new air fluid level in the pneumonectomized pleural space. Occasionally this is all that is required to drain the space and allow the defect to close.
- Failure to close and contamination of the pleural space requires:
  - Creation of a chest wall window and primary closure of the BPF.
  - Serial debridements of the infected space with either continuous packing or wound vacuum dressing.
  - Filling of space either with space-occupying tissue (omentum, muscle) or antibiotic solution and primary closure of chest wall (*Clagett procedure*) 📖 p707.
  - Early fistulas are usually associated with technical difficulty at surgery, repeat thoracotomy and resuturing of stump is indicated.
  - Fistulae occurring >2 weeks postoperatively are usually associated with empyema: iv antibiotics should be administered.

Empyema

Key facts

- Defined as ***pus in the pleural space***: often from the lung parenchyma or translocated through the chest wall, mediastinum, or abdomen.
- An empyema may grow organisms or it may be sterile (1/3 of cases), depending how long the patient has been on antibiotics.
- The aim of treatment is to drain and sterilize the space, and re-expand the lung which can done using VATS with limited decortication in early stage empyemas, but in recurrent empyema ± bronchopleural fistula (📖 p702) may require a chest wall window for chronic debridement.

Pathology

- A ***parapneumonic pleural effusion*** is common. The majority resolve with antibiotic therapy. ***Postpneumonic empyema*** forms when bacteria infect the pleural fluid or pleural space. It is the commonest cause of an empyema.
- ***Postsurgical empyema*** accounts for 20% of cases of empyema and can follow surgery to the mediastinum, esophagus, and lungs, most commonly following pneumonectomy. It can also follow surgery or injury to the abdomen. Presentation is usually in the early postoperative period, but it may develop several years after surgery.
- Empyema may also follow rupture of a lung abscess or infected pleural bleb, inhaled foreign bodies or bronchopleural fistula.
- Empyema may also result form thoracic trauma, ruptured esophagus, pericarditis, abdominal processes such as cholangitis and diverticulitis with translocation of bacteria across the diaphragm, mediastinitis, chest wall or spine osteomyelitis, rupture of lung abscess or infected pleural bleb, and inhaled foreign body.

Stage I: acute exudative phase

- Typically occurs 2–5 days after a pneumonia.
- Accumulation of fluid with low cellular content and viscosity.
- Characterized by low WBC, LDH, glucose, and a normal pH.
- Can be successfully treated with antibiotics only.

Stage II: fibrinopurulent phase

- Typically occurs 5–14 days after a pneumonia.
- Turbid or purulent fluid with heavy fibrin deposits.
- Appearance of simple loculations and septations.
- May have bacterial invasions, and high numbers of polymorphonuclear leukocytes (PMNs) and lymphocytes.
- Characterized by low pH, glucose, and increased LDH.
- Antibiotics and chest tube drainage is required, ± VATS decortication.

Stage III: chronic organizing phase

- Lung trapping by collagen, visceral and parietal pleural peel with ingrowth of fibroblast and capillaries.
- Antibiotics and aggressive decortications, generally by thoracotomy.
- Bacteriology.

- Includes staphylococcus, *Haemophilus influenza*, Gram –ve bacilli, *Streptococcus pneumonia*, *Streptococcus pyogens*, *Mycoplasma*.
- Empyema that has been present for >2 weeks has a higher risk of conversion from VATS to open for decortication.

Presentation

Patients may complain of fever, productive cough, dyspnea, chest wall pain, malaise, fatigue. Physical examination may reveal decreased ipsilateral chest wall expansion, egophony, dullness to percussion, chest wall tenderness. Complications include ***empyema necessitans*** (discharge of empyema through the chest wall), pulmonary fibrosis, chest wall contractures, osteomyelitis, pericarditis, subphrenic, or mediastinal abscesses.

Investigation

- CXR, PA and lateral, and/or decubitus films to determine loculation.
- Chest CT: provides anatomy, and helps planning for chest tube placement of surgery, as well as showing thickened pleural peel, and presence of calcifications in the peel which help determine if thoracotomy rather than VATS necessary.
- Thoracentesis: Gram's stain and culture, ***Light's criteria*** distinguishes transudative effusion from exudative:
  - Exudative if specific gravity is >1.02, protein content >2.0 g/dL, and at least one of the following: ratio of pleural fluid protein to serum >0.5, ratio of pleural fluid LDH and serum >0.6, total pleural fluid LDH >2/3 upper limit of normal, ratio of pleural fluid albumin and blood <1.

Management

Stage I

- Sensitivity-appropriate antibiotics.

Stage II

- Chest tube drainage (small or large bore).
- Thrombolytics most likely to be effective during early stage empyema, and should be reserved for patients who are poor surgical candidates.

Stage III and failure of stage II to resolve

- If pleural space drainage is ineffective, or the effusion has loculated appearance on imaging, a ***VATS decortication*** should be performed without delay
- If complete lung expansion is not achieved by VATS, then should convert to open thoracotomy: in processes that have been going >14 days, or in patients who have chest wall tenderness, need for conversion for open thoracotomy is higher.
- *Open decortication* (📖 p706) indicated for late stage II, stage III, and incomplete lung expansion with VATS.
- *Chest wall window* (Eloesser flap) (📖 p707) may be required if lung cannot expand and empyema chronically reaccumulates, or in the case of a BPF that cannot be closed.

Decortication

Key facts

- This is a major undertaking usually performed through a posterolateral thoracotomy attempting to completely remove what is usually densely adherent of visceral peel off the lung to allow it to completely re-expand.
- It is indicated for treatment of stage III empyemas: early stage II decortications are difficult because there is less of a plane between the lung and the peel than in later stage III empyemas.

Procedure

Set-up

- General ET anesthesia with invasive arterial monitoring, and usually a central line.
- A flexible bronchoscopy (📖 p654) is performed via a single lumen tube, before lung isolation is obtained using a bronchial blocker or a double lumen tube,
- The patient is then positioned in the *lateral, decubitus position* (📖 p39) with spine parallel to and close to the edge of the table, using sandbags, or rolls and tape, usually with the upper arm positioned on a rest, the lower arm flexed in front of the head.

Procedure

- The aim is complete removal of the rind, leaving the visceral pleura intact allowing complete expansion of the lung.
- Entry into the pleural space is not like a normal thoracotomy: once through the intercostal muscles the first thing encountered may be a thick, cream-coloured rind, and there may be no plane between this and the lung parenchyma.
- In order to safely get enough space to place a retractor and spread the ribs it is usually safest to start with a local pleurectomy, carefully wiping the rind off the chest wall with finger tips. This leaves raw surface area which should be cauterized.
- Once a retractor can be safely introduced the correct plane to elevate the rind off the lung parenchyma should be found: it's easiest to do this cutting carefully into the rind with a 15 or a 23 blade with the lung inflated.
- As soon as the plane is identified then counter-traction on the lung and traction on the rind, with careful blunt dissection using peanuts or multiple blades and or/a periosteal elevator, similarly usually permits most of the rind to be removed without damaging the lung parenchyma.
- The most difficult areas are the fissures (although air leaks here usually seal up postoperatively) and along the diaphragm.
- Careful hemostasis to avoid retained hemothorax and reinfection of space is mandatory.

Postoperative management

- Lung expansion is encouraged by leaving chest tubes on water seal suction.
- Chest tubes should remain until output is minimal.
- Excessive bleeding from raw surfaces is an occasional problem that may require re-exploration, whereas air leaks often resolve quickly due to the sticky nature of the pleural space.

Eloesser window

- This is used in the setting of chronic recurrent empyema, including after pneumonectomy, and it allows chronic drainage to the skin.
- It involves removal of ribs over an accessible location (ideally in a dependent posterior area to improve drainage) with preservation of muscle, soft tissue, and skin.
- 1/3 of scapula may need to be removed with a sternal saw to avoid locking of the scapula tip in the window.
- Muscle and skin is marsupialized by sewing to parietal pleura of chest wall (Figure 14.3).
- After serial debridements, when the space is sterile and any BPF has resolved, the space may be closed by filling with antibiotic solution (***Clagett procedure***) or omentum or muscle and the marsupialized tissue is released and used to close the surface of the wound.
- A stoma bag can be placed over the site.

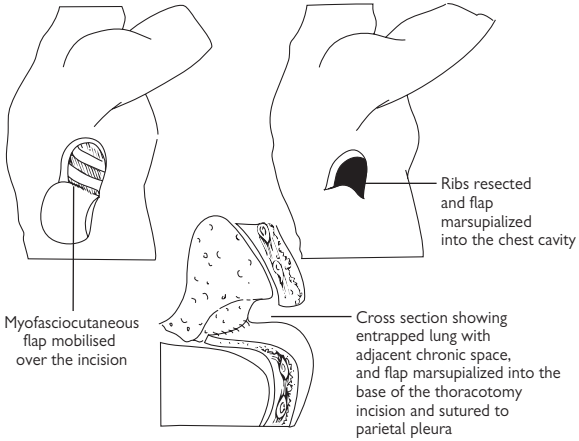


Fig. 14.3 Eloesser window marsupialization of chest wall tissue.

Fungal lung disease

Key points

Pathogens include histoplasmosis, coccidioidomycosis, blastomycosis, and opportunistic infections, e.g., aspergillosis, cryptococcosis.

Histoplasmosis

- Endemic in midwestern US states such as Michigan, Ohio, and Indiana.
- ***Histology:*** budding yeasts consistent with *Histoplasma capsulatum*.
- ***Presentation:*** fever, malaise, pleuritic chest pain, flu-like symptoms, but often asymptomatic. Can also lead to insidious ***mediastinal fibrosis, broncholiths, traction diverticulum*** of the esophagus.
- ***Investigation:*** sputum culture, CXR, chest CT which show solitary pulmonary nodules, calcified nodules, and lymph nodes.
- ***Treatment:*** if symptomatic, amphotericin B.
- Surgery for medical refractory disease, symptomatic broncholiths, traction esophageal diverticulum, or mediastinal fibrosis resulting in destruction of lung tissue.
- Surgery can have high mortality or catastrophic bleeding: consider cardiopulmonary bypass or stand by for difficult hilar disease.

Coccidioidomycosis

- Infection from inhalation of *Coccidioides immitis* spores. Also called ***valley fever***. Endemic to western and southwestern USA and Mexico.
- ***Histology:*** demonstrates spherule containing endospores.
- ***Presentation:*** fever, cough, fatigue, dyspnea (valley fever), arthralgias, erythema nodosum, rarely progressive pneumonia, cavitary lesions, thin wall pulmonary cavities.
- Extrapulmonary complications includes CNS infections.
- ***Investigation:*** serum testing for IgM or IgG antibodies, sputum culture, CXR and chest CT, and bronchoscopy with BAL.
- ***Treatment:*** most infections self-limiting—amphotericin B for severe disease, fluconazole or itraconazole for 3–6 months.
- Indications for surgery include failure of medical management with persistent cavitary lesion, hemoptysis, secondary infection, or BPF.

Blastomycosis

- Caused by *Blastomyces dermatidis*. Endemic to southeastern USA, as well as the midwest. Results in suppurative granulomatous pulmonary disease.
- ***Histology:*** broad-based budding yeasts and neutrophil infiltration.
- ***Presentation:*** flu-like symptoms, myalgias, non-productive cough, acute pneumonitis, ARDS, chronic pyogranulomatous lung disease, with extrapulmonary symptoms, e.g., skin ulcers and multiorgan dissemination.
- ***Investigation:*** serodiagnosis unreliable, sputum cultures, bronchoscopy with BAL, CXR, and chest CT.
- ***Treatment:*** most infections self-limiting, amphotericin B for severe disease, fluconazole or itraconazole for 6–12 months.
- Indications for surgery include failure of medical management with persistent cavitary lesion, hemoptysis, secondary infection, or BPF.

Aspergillosis

- *Aspergillus fumigatus* is an opportunistic fungus that is found in soil and is inhaled. Especially in patients with AIDS, immunosuppression including prolonged periods of neutropenia, and chronic steroid use.
- ***Presentation:*** fever, cough dyspnea, hemoptysis, anorexia, malaise, may be complicated by aspergilloma, chronic necrotizing aspergillosis, invasive pulmonary aspergillosis.
- ***Investigation:*** sputum cultures (negative in 50% of cases), bronchoscopy with BAL, CXR, and chest CT.
- ***Treatment:*** IV voriconazole for invasive pulmonary aspergillosis, with amphotericin B as an alternative therapy for 6–12 weeks, with subsequent maintenance therapy, voriconazole for necrotizing aspergillosis for 6 months, amphotericin B for severe disease.
- Indications for surgery: asymptomatic aspergilloma can be left alone. Anatomic resection for aspergilloma, persistent hemoptysis, BPF, invasive disease, potential invasion of great vessels or pericardium.

Mucormycosis

- Opportunistic fungal infection caused by *Mucor*, *Rhizopus* and other zygomycetes. Found in soil. Can affect lungs in patients with AIDS, transplant patients, and other immunosuppressed patients.
- ***Histology:*** non-septate hyphae, right angle branching, vascular invasion.
- ***Presentation:*** bronchopneumonia, hemoptysis, lung infarction, chest wall invasion.
- ***Investigation:*** sputum cultures are not sensitive, CXR and chest CT.
- Tissue diagnosis demonstrating right angle branching non-septate hyphae is the gold standard.
- ***Treatment:*** amphotericin B for aggressive disease, long-term to lifetime posaconazole therapy for less aggressive disease.
- Surgery: anatomic resection of lung and adjacent tissue followed by long-term to lifetime posaconazole therapy.

Cryptococcosis

- Caused by the encapsulated yeast *Cryptococcus neoformans*. In addition to pulmonary disease, also associated with CNS disease, specifically meningitis. Commonly found in pigeon droppings.
- ***Histology:*** by budding yeasts.
- ***Presentation:*** cough, low-grade fever, fatigue, meningitis, pulmonary masses, and mediastinal lymphadenopathy.
- ***Investigation:*** serology for cryptococcal antigen is sensitive, CSF should always be examined, sputum cultures, bronchoscopy with BAL, CXR, and chest CT.
- ***Treatment:*** medical therapy is the gold standard with amphotericin B for aggressive disease including meningitis, fluconazole for less aggressive disease for 3–12 months.
- ***Surgery:*** mediastinoscopy to diagnosis suspicious mediastinal lymphadenopathy for diagnosis, wedge resection of suspicious lung nodules for diagnosis.

Other lung infections

Lung abscess

Etiology

- *Primary:* direct infection via aspiration, esophageal disease, necrotizing pneumonia, primary tuberculosis, or fungal infections.
- *Secondary:* bronchial obstruction, malignancy, hematogenous bacterial spread, such as endocarditis, cystic disease, collagen vascular disease such as Wegner granulomatosis.

Presentation

- Productive cough, fever, rigors, fatigue, malaise, pleuritic chest pain, dyspnea, hemoptysis.
- Absence of fever or leucocytosis may suggest non-infectious causes of abscess, such as malignancy.

Investigation and management

- *Investigation:* sputum and blood cultures, CXR and chest CT, bronchoscopy for BAL.
- Avoid FNA, as may contaminate the pleural space and result in empyema.
- Medical management is 6–8 weeks appropriate antibiotics: 90% of patients respond to targeted antibiotic therapy alone and do not require surgical intervention, and aggressive chest physiotherapy.
- 1/3 of percutaneous drainage attempts fail, and should be avoided if possible.
- *Indications for surgery:*
  - Failed medical therapy.
  - Persistent endobronchial obstruction.
  - Formation of an empyema.
  - Hemorrhage.
  - Bronchopleural fistula.
  - Inability to rule out malignancy.

Surgical options

- Anatomic lobectomy.
- Pneumonectomy (avoid where possible for benign disease).
- All bronchial stumps should be covered with viable tissue.
- Intercostal muscle, omentum, pleural flap, pericardial fat pad.
- Cavernostomy.
- Thoracoplasty is rarely required to obliterate an apical space.

Hydatid pulmonary disease

- Caused by tapeworm infestation of *Echinococcus*. Results of invasion of lung and/or rupture of hydatid cyst into the pleural space from adjacent structures (such as transdiaphragmatic from the liver).
- ***Presentation:*** cough, dyspnea, hemoptysis, expectoration of cyst material, acute pleural drainage can cause pain and ***anaphylactic shock.***
- *Diagnosis:* serology (IgG, IgM, hemagglutination), Casoni skin test, CXR and chest CT.
- ***Treatment:*** albendazole is drug of choice (3–6 months).

- ***Surgery:*** avoid spillage. Cystectomy ± anatomic resection, may require concurrent liver resection or cystotomy and capitonnage, concurrent albendazole therapy.

Amoebiasis

- Parasitic infection with *Entamoeba histolytica*. Transmitted via feco-oral route forming ***liver abscess***, which ruptures or extends into surrounding tissues, producing ***pleuro-pulmonary contamination***.
- ***Presentation:*** right-sided chest pain, pain radiating to shoulder, cough initially unproductive becoming productive of purulent chocolate brown sputum, fever, malaise, anorexia and weight loss, dyspnea. Parasites found in sputum.
- ***Investigation:*** sputum culture.
- ***Treatment:*** metronidazole, drainage of pleural fluid.

Actinomycosis

- Gram +ve actinomyces.
- ***Presentation:*** chest pain, dyspnea, hemoptysis, cough, fever, weight loss, invasion of chest wall with drainage.
- ***Investigation:*** wound and sputum culture.
- ***Treatment:*** penicillin for 5 weeks, drainage of pleural fluid.

Nocardiosis

- Aerobic Gram +ve organism. *Norcardia asteroides*.
- Presentation: cough, fever, malaise.
- Investigation: culture of sputum or pleural fluid.
- Treatment: Sulfonamide therapy, drainage of pleural fluid.

Interstitial lung disease

Key points

- Role of surgery is for tissue diagnosis.
- Lung biopsy helps determine diagnosis and mode of therapy (i.e., corticosteroids or lung transplantation).
- Severely diseased or 'burned out' tissue should be avoided for biopsy as histological architecture may be distorted.
- If both sides are equally diseased, VATS biopsy is performed on the left, as right lung is better equipped to tolerate single-lung ventilation.
- Open biopsy on ventilated lung should be performed on patients with severe respiratory distress.

Pathology

- Diffuse inflammation and fibrosis of the lung parenchyma characterized by inflammatory cells, fibroblastic foci, collagen accumulation.
- The underlying aetiology include a number of conditions (Table 14.5). as well as collagen vascular disease, SLE, rheumatoid arthritis, and environmental exposure (silicosis, berylliosis), drug-induced (smoking, amiodarone, bleomycin), granulomatous disease, sarcoidosis, Wegener granulomatosis, idiopathic interstitial pneumonia, TB, lymphangiomyotosis, Langerhans cell histiocytosis.

Investigation

- Chest CT: reticular pattern, ground glass infiltrates, honeycombing
- Pulmonary function tests: restrictive lung disease pattern, reduced DLCO, TLC, FRC and RV, while FEV_1, and FVC may be normal.
- Transbronchial needle biopsy has low yield, low accuracy.

Lung biopsy

- Lung biopsy is the gold standard and provides the most accurate diagnosis. Indications for lung biopsy include no definitive diagnosis before starting steroid therapy, need to exclude neoplasm infection.
- Goals are to biopsy areas of parenchymal that both appear normal and abnormal on CT, avoiding severely diseased or 'burned out' areas of parenchyma that have severe amount of fibrosis such as lingula and middle lobe which often have the highest areas of 'burned out' disease.
- 2–3 biopsy specimens, specimen $\geq 3 \times 2 \times 1 cm^3$ in size.
- Open biopsy: in critically ill patients who would not tolerate single-lung ventilation (can be done in ICU via an anterior thoracotomy).
- VATS: procedure of choice for elective lung biopsy.
- Accuracy and outcomes equivalent to open lung biopsy, 5% 60-day mortality, 20% morbidity.
- If both sides are equally diseased, then choose left side for biopsy, as the right lung is best equipped to tolerate single-lung ventilation.

Complications of surgical lung biopsy

- Exacerbation of underlying lung disease: avoid by limiting FiO_2 level to avoid O_2 toxicity and free radical production.
- Limit barotrauma and excessive tidal volumes, and use staple line reinforcement to avoid prolonged air leak.

Table 14.5

Etiology	Presentation	CT findings	Histology	Treatment	Survival
AIP	Acute onset respiratory failure mimicking ARDS	Ground glass opacities with areas of consolidation	Proliferative phase: Hyaline membranes, diffuse alveolar injury Organizing phase: Abundant type II pneumocytes and fibroblasts	Supportive therapy with mechanical ventilation	Poor survival
COP	Subacute onset with short duration of symptoms. Associated with infectious pneumonia	Areas of consolidation, small nodules and subpleural and/or peribronchial ground glass attenuation	Patch areas of organizing pneumonia	High-dose steroids	Usually good recovery after 4–6 week course of steroids
DIP	Occurs in heavy smokers and patients with environmental or drug exposures	Lower lobe predominant ground glass attenuation and consolidation	Diffuse distribution of alveolar macrophages	High-dose steroids	Good prognosis after steroid therapy
LIP	Slow progression of dyspnea and cough	Ground glass attenuation, interstitial thickening, and pulmonary nodules	Diffuse interstitial lymphoid hyperplasia	Steroids	Good prognosis after steroid therapy, but may progress to lymphoma

(*continued*)

Table 14.5 (*Continued*)

Etiology	Presentation	CT findings	Histology	Treatment	Survival
NSIP	Derivative of UIP, but milder symptoms. Progressive dyspnea and cough	Reticular markings with bronchiectasis, ground glass attenuation, and volume loss	Diffuse fibrosis and inflammation	Steroids, cyclophosphamide, colchicines, azathioprine, IFN-γ1b	Median survival of 56 months
RBILD	Occurs in heavy smokers. Progressive dyspnea	Ground glass attenuation with centrilobular nodules	Centriacinar proliferation of alveolar macrophages in respiratory bronchioles	Smoking cessation, steroids	Usually reversible after smoking cessation
UIP	Progressing cough, dyspnea, and reduced vital capacity	Patch reticular fibrosis pattern and honeycombing	Fibrosis with without fibroblastic proliferation	Steroids, cyclophosphamide, colchicines, azathioprine, IFN-γ1b	Median survival of 33 months

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Hemoptysis

Key facts

- Hemoptysis is expectoration of (coughing up) blood, either mild or catastrophic: minor is streaks in sputum, moderate <400mL/24h, massive >400mL/24h, exsanguinating>1L.
- In ***massive hemoptysis*** patients generally do not die from exsanguination, but 'drowning' of unaffected lung because of inability to secure the airway.
- Consider bronchoscopic-guided selective intubation to protect the unaffected lung, with coil embolization to stop bleeding.
- Definitive surgery often requires thoracotomy and anatomic resection.

Pathology

Broad range of causes of hemoptysis:

- Bronchiectasis.
- Infectious, e.g., TB, fungal infection, pulmonary abscess.
- Malignant, including primary and secondary tumors of the lung and airway.
- Trauma, usually penetrating airway injury.
- Iatrogenic: chemotherapeutics such as bevacizumab (Avastin®), TIF after tracheostomy, pulmonary artery rupture from PA catheter.
- Other causes include cystic fibrosis, congenital atriovenous malformations, Wegener granulomatosis, Goodpasture syndrome.

Investigation

- CBC, PT, PTT, INR, DIC panel including fibrinogen, Chem 7.
- If stable, chest CT angiogram to determine relevant anatomy and extravasation.
- Bronchoscopy either flexible or rigid can identify which lung is the source of hemoptysis, and allow for selective intubation of non-involved lung.
- Rigid bronchoscopy is advantageous as it allows for concurrent ventilation and large tools for evacuating clot.
- Bronchial artery angiogram can be diagnostic and therapeutic, permitting coil embolization may stop or temporize bleeding.

Therapy

- Correction of all coagulopathy.
- Administration of antifibrinolytics such as factor VIIa.
- Bronchial artery embolization.
- Often indicated for fungal aetiology such as aspergillosis.
- Bronchoscopic techniques for localized area of hemoptysis.
- Iced saline lavage.
- Epinephrine or phenylephrine lavage.
- Cryoprobe.
- Argon beam coagulation.
- Laser coagulation.

Indications for surgery

- Failure of medical management in medical diseases and localization of disease.
- TIF.
- Congenital abnormalities such as known pulmonary artery aneurysms or AVM.
- Failed bronchial artery embolization.

Contraindications for surgery

- Generalized source of hemoptysis—consider ECMO and treatment of underlying lung disease.
- Palliative condition.
- Metastatic end-stage malignancy.

Surgical technique

- Thoracotomy.
- Anatomic resection: lobectomy or pneumonectomy.

Congenital disorders of the lung

Key facts

- Many of the disorders of the lung can be diagnosed *in utero*.
- All should be part of the differential diagnosis for neonatal respiratory distress and cyanosis.
- Most are treated with anatomical resection.

Congenital lobar emphysema

- ***Pathology:*** this is over-distension of one or more lobes of the lung, leading to life-threatening compression of the unaffected lung and mediastinal shift. The left upper and right middle lobes are the most commonly affected.
- Due to dysfunctional cartilage development of involved bronchus, resulting in air trapping, associated with congenital cardiac anomalies.
- ***Presentation:*** usually affects newborns and infants, and is diagnosed in the first 6 months of life due to life-threatening conditions including respiratory distress and cyanosis. Less acute symptoms include dyspnea, wheezing, cough and tachypnea. Late symptoms include recurrent pneumonia, failure to thrive, and chest wall deformity.
- ***Investigation:*** CXR demonstrates hyperinflation of the affected lobe, with contralateral mediastinal shift. Paucity of lung markings on the ipsilateral side. Chest CT usually diagnostic.
- Surgical management consists of anatomic lobectomy.

Pulmonary sequestration

- ***Pathology:*** congenital mass of non-functioning lung tissue that has no communication with the tracheobronchial tree, and derives its blood supply from systemic artery system (thoracic aorta, the abdominal aorta travelling via the inferior pulmonary ligament, or the intercostal arteries), not the bronchial arterial circulation. Two types: venous drainage for *intralobar* (most common) is the pulmonary venous system, while venous drainage for *extralobar* is the systemic circulation.
- ***Histology:*** bronchiectasis and vascular hypertrophy.
- ***Presentation:*** cyanosis, respiratory distress in younger patients, in older patients may result in recurrent infections, empyema, chronic pleuritic chest pain. Extralobar sequestrations can be associated with congenital heart defects, vertebral defects, and diaphragmatic abnormalities. Extralobar has a 3:1 male to female ratio.
- ***Investigation:*** chest CT angiogram, interventional angiogram can identify anomalous arterial supply, and at the same time coil embolize subdiaphragmatic feeding arteries to limit risk of resection (Fig. 14.4).
- ***Management:*** surgical resection (VATS or open). Failure to identify feeding artery can result in retraction beneath diaphragm and catastrophic exsanguination: pre-op coil embolization of feeding artery mitigates the this risk.
- Identify and ligate feeding artery often in the inferior pulmonary ligament. Only sublobar resection needed. Ventilation of lung will identify demarcation of non-ventilated sequestration and normal lung tissue.

Congenital cystic adenomatoid malformation

- ***Pathology:*** hamartomatous type lesion with overgrowth of bronchioles with increased terminal respiratory structures that are connected to the tracheobronchial tree.
- Histology: cyst-like bronchioles lined with either pseudostratified columnar or cuboidal epithelium. Absence of cartilage in the cyst wall.
- Results from bronchial atresia with subsequent dysplastic lung growth beyond the atretic segment. Lesions occur on either side, most commonly in the lower lobes. They are classified into three types:
  - Type I (70%): large cystic spaces confined to one lobe.
  - Type II (20%): numerous small cysts, <2cm in size.
  - Type III (10%): multiple small cysts, <0.5cm in size, with large amount of solid adenomatoid hyperplasia or bronchial structures.
- ***Presentation:*** in newborns with respiratory distress, cyanosis, dyspnea, tachypnea, pneumothorax, poor feeding, pulmonary hypoplasia. Older children and adults may have milder symptoms or are asymptomatic.
- Investigation: most are diagnosed *in utero* by ultrasound. Newborn plain babygram may reveal cystic or solid mass, with contralateral mediastinal shift. Cysts may contain air fluid level.
- CT scan can distinguish from congenital diaphragmatic hernia.
- ***Management:*** some regress spontaneously, most require anatomic lobectomy. Type I: good prognosis. Type II and III: 50% mortality.

Bronchogenic cysts

- ***Pathology:*** abnormal non-communicating bronchogenic tissue that may be intrapulmonary or mediastinal. Represents 10–15% of mediastinal masses in children. In children, 65% are located in the mediastinum.
- Derived from the embryonic foregut.
- ***Histology:*** squamous or ciliated columnar epithelium. They may also contain hyaline cartilage, smooth muscle cells, and glands.
- Cysts occurring early in gestation tend to be centrally located, while those developing late in gestation tend to be peripheral.
- ***Presentation:*** respiratory distress from compression symptoms especially in newborns. Dysphagia from extrinsic compression of the esophagus, recurrent infections, fever, pain, dyspnea, anorexia. Often are asymptomatic and found incidentally.
- ***Investigation:*** CXR and chest CT. FNA is not necessary.
- ***Management:*** if surgical excision is not possible, than the cyst is opened and the lining excised and fulgurated to prevent recurrence.

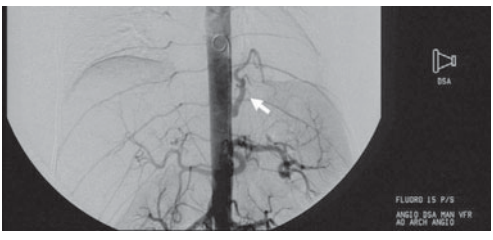


Fig. 14.4 Subdiaphragmatic feeding artery in left intralobar pulmonary sequestration.

Pulmonary tuberculosis 1

Key facts

- Airborne infection caused by *Mycobacterium tuberculosis*, more rarely *M. bovis* or *M. africanum*, transmitted as aerosol, or in the case of *M. bovis* by ingestion of contaminated milk.
- Increased incidence in patients with HIV infection. Low socioeconomical status, substance abuse, immigrants, inhabitants of nursing homes and prisons.
- Surgery is indicated for failure of medical therapy to remove destroyed lung, deal with secondary infections, e.g., *Aspergillus*, and complications such as massive hemoptysis and bronchopleural fistula.
- Although 2 billion patients worldwide are infected with TB, <15 000 cases annually are diagnosed in the USA: the numbers of operations for TB in the west are correspondingly very limited in number.

Pathology

- The characteristic lesion is ***tuberculous granulomas*** with ***central caseation***. The primary infection is characterized by a ***Ghon focus***, usually at the apex of the upper lobe.
- Later reactivation results in growth of mycobacterium within the liquefied caseum. Bronchial involvement results in expectoration of infected material. Seeding via the bloodstream also occurs.
- Following primary infection healing a ***dormant phase*** follows, which may last many decades. During this phase the patients are asymptomatic and the only evidence of infection is radiographical visualization of the Ghon focus and a ***positive tuberculin skin test***.
- There is a 10–15% reactivation rate, usually involving the lungs.

Presentation

Presentation is extremely variable, but classically symptoms include low-grade fever, weight loss, night sweats, malaise, productive cough, hemo-ptysis, chest pain, and dyspnea. Complications include destroyed or cavitated lung, bronchostenosis with middle lobe atrophy (Lady Windermere or middle lobe syndrome), bronchopleural fistula, and secondary infection with aspergillosis.

Investigation

- For definitive diagnosis one of the described *Myocbacteria* species must be cultured, most commonly from sputum (which can be obtained by bronchoalveolar lavage, bronchial washings), but also from pus, CSF or biopsied tissue (FNA, mediastinoscopy, etc.).
- For detection of acid-fast bacilli, fluorescence microscopy is more sensitive than conventional Zieh–Nielson staining, confirmed by PCR.
- CXR may show consolidation or cavities in the upper lobes, with mediastinal and hilar lymphadenopathy and pleural effusions in active pulmonary TB. Miliary TB described as many sub-cm nodules throughout both lungs.
- CT is used to identify parenchymal destruction, caseous granulomas, Ghon foci, lymphadenopathy.

- A +ve tuberculin skin test (***Mantoux test***) of more than 5mm induration indicates TB infection, except in people previously immunized with the BCG vaccine where there is a high number of false positives. Thoracentesis for pleural effusion: pleural fluid analysis may demonstrate protein >5.0gm/dL, 50% lymphocytes, high ednosine deaminase level.

Management

Medical

- Avoid chest tubes in TB pleural effusion as high rate of superinfection and they generally resolve with appropriate antibiotic therapy.
- 2–3 months of triple therapy (isoniazid, rifampin, and pyrazinamide) followed by 4 months of dual therapy (isoniazid and rifampin).

Two months of triple therapy (isoniazid, rifampicin, and pyrazinamide) followed by 4 months of dual therapy (isoniazid and rifampicin). Drug resistance may require alternative drugs such as ethambutol or streptomycin. Multidrug resistant TB is often responsible for lesions in patients requiring surgery. All surgical patients should be on antituberculous medical therapy for 3 months preoperatively, and undergo aggressive nutrition and physiotherapy to correct their anabolic state where clinical presentation permits.

Indications for surgery

Surgery is indicated for failure of medical therapy to remove destroyed lung, deal with secondary infections, e.g., *Aspergillus*, and complications such as massive hemoptysis, bronchopleural fistula, bronchostenosis, and re-expansion of trapped lung.

Pulmonary tuberculosis 2

Surgery

The aim of surgery is complete resection of diseased lung tissue along with conversion of sputum. Should be complemented with neo- and adjuvant antituberculous multidrug therapy. Surgery may be hazardous due to previous infection/surgery. May require extrapleural lobectomy or pneumonectomy. Care is needed not to contaminate pleural space.

- General ET anesthesia, invasive arterial and central venous monitoring, bronchoscopy, and lung isolation.
- Patients is positioned in the lateral decubitus position.
- *Posterolateral thoracotomy:* 5th intercostal space incorporating bronchopleural fistulas or prior Eloesser flaps, preserving latissimus dorsi for a muscle flap to cover the bronchial stump, and preserving serratus to close the thoracotomy wound (and reduce the risk of winging the scapula, which would compromise wound healing posteriorly).
- Omentum may also be mobilized to cover the bronchial stump.
- Rib resection is necessary in patients with hemithorax contracture.
- Dense adhesions between the parietal and visceral pleura mean that an *extrapleural dissection* is needed to mobilize the lung, taking great care to avoid the aorta on the left side, and the azygos, IVC, and esophagus on the right side, and the subclavian vessels at the apex.
- Lobectomy and pneumonectomy are performed in the standard fashion after mobilizing the lung: occasionally dense adhesions at the hilum necessitate intrapericardial division of vessels.
- The pleural cavity is irrigated with antibiotic solution, and the thoracotomy closed in layers with multiple absorbable sutures, and bronchoscopy performed to remove any secretions.

Outcomes

- Operative mortality 3–5%, sputum conversion 95%.
- Complications include empyema, late bronchopleural fistula—this is managed with *Eloesser window* (📖 p705) until the space is sterilized, and then the stump dehiscence is covered with a muscle flap, omentum or a *thoracoplasty* (where the chest wall is partially resected so that it collapses into effectively obliterating the pleural space.
- Bleeding, infection of the pleural space with TB empyema, BPF, acute lung injury.
- 95% sputum conversion.

Multidrug resistant tuberculosis (MDR-TB)

- Defined as resistance to both rifampin and isoniazid.
- Medical treatment involves addition of a fluoroquinolone such as levofloxacin and amikacin.
- Indications for surgery include refractory to medical therapy, localized disease or persistent cavity, complications such as hemoptysis.
- Outcomes: 93% disease free-outcome.

Non-tuberculosis mycobacteria (NTM)

- Defined as new and old species other than TB that affect public health, such as Mycobacterium avium, Mycobacterium bovis, and Mycobacterium africanum.
- Medical treatment includes isoniazid, rifampin, ethambutol, with addition of a macrolide and a fluoroquinolone if resistant to any of the described regimen.
- Indications for surgery include disease refractory to medical therapy, localized disease or persistent cavity, complications such as hemoptysis.

Bronchiectasis

Key facts

- Bronchiectasis is irreversible dilatation of the bronchi usually associated with inflammation.
- It is caused by the destruction of the elastic and muscular components of the bronchial wall and is either a localized or diffuse process.

Pathology

Three main types are described: ***cylindrical***—often seen with TB, ***varicose***, and ***saccular***—usually seen after infection or obstruction.

Congenital

Congenital bronchiectasis usually results in diffuse disease (the exception being cystic fibrosis which tends to primarily affect the upper lobes). Congenital bronchiectasis may be due to a congenital immunoglobulin deficiency, cystic fibrosis, ciliary dysmotility disorders, or connective tissue disorders.

Acquired

- Bronchiectasis secondary to tuberculous infection usually affects the upper lobe and that secondary to any other bacterial or viral chest infection affects the lower lobes. More common causative organisms include adenovirus, influenza virus, HIV, *Staphylococcus aureus*, *Haemophilus influenzae*, anaerobes, *Mycobacterium tuberculosis*.
- Chronic external compression of a bronchus, due to tumor mass (carcinoid more common than malignancy) or enlarged lymph nodes.
- Gastric aspiration, pneumonia secondary to foreign body aspiration and inhalational injury, e.g., ammonia.
- Allergic bronchopulmonary aspergillosis.
- Other diseases: rheumatoid arthritis, ulcerative colitis, Sjögren syndrome.

Presentation

Recurrent productive cough. Sputum is purulent. More advanced disease may present with hemoptysis. If disease affects the upper lobes the cough may be non-productive.

Investigations

- CXR: may be normal or may show cystic lesions with air–fluid levels. Bronchial wall thickening may be evident.
- CT scan: high-resolution scans routinely used to demonstrate the dilated airways.
- *Sputum culture:* pathogens often identified. Commonly *H. influenzae*, *P. aeruginosa*, *S. pneumoniae*.

Management

The aim of therapy is to treat any infection, any underlying cause for bronchiectasis, optimize clearance of secretions, and reduce air flow limitation with bronchodilators. Physiotherapy is very important, using postural drainage techniques, chest percussion, and vibration. Aggressive antibiotic therapy for acute exacerbations prolongs periods of remission.

Surgical options

Lung resection, e.g., lobectomy (📖 pp680–689) or segmentectomy (📖 p694) is indicated in those patients with sufficient pulmonary reserve who remain symptomatic despite optimal medical therapy. Massive hemoptysis may also necessitate surgical resection of the affected lung. The aim is to preserve as much normal lung parenchyma as possible, hence surgical candidates tend to be those with limited disease. In those with insufficient pulmonary reserve embolization may be an option but is associated with rebleeding.

Emphysema

Key facts

- Pulmonary emphysema results from the destruction of air spaces, with a loss of elastic recoil of the lungs and alveolar capillary surface for gas exchange.
- 4 million Americans are currently affected by emphysema.
- The only medical therapy that has shown to impact survival is O_2 therapy: surgery is indicated for selected patients.

Pathology

Destruction of pulmonary parenchyma reduces functional lung mass, resulting in impaired gas exchange; as well as loss of elastic recoil which leads to hyperinflation (bullae), loss of the mechanical advantages seen in normal lung volumes, and compression of relatively normal areas.

Presentation

Patients present with insidious onset of dyspnea. Chronic productive cough is common. Cyanosis. May present with acute infective exacerbation, pneumothorax. Smoking history is common. Physical examination reveals characteristic barrel chest, widened intercostal spaces.

Investigation

- ***CXR:*** large bullae may be visible, flattened diaphragm, reduction in lung markings.
- ***CT chest:*** destruction of lung parenchyma, bullous disease.
- ***Lung function:*** increased residual capacity, decreased DLCO.

Rationale for lung volume reduction surgery

Medical management is supportive, (smoking cessation, physiotherapy, bronchodilators and steroids, home O_2) and associated with poor functional and prognostic outcomes. Rationale for surgery is that removal of hyperinflated and underperfused lung parenchyma will allow expansion of functional ventilated and perfused lung, restoring mechanical advantage by normalizing lung volume and diminishing dyspnea.

- 1957 Brantigan and Mueller[1] first described excision of hyperinflated lung parenchyma with 18% mortality.
- 1996 Cooper *et al.*[2] reported on 150 patients undergoing LVRS: 2.5% mortality, 14.4% mortality at 3 months, and 23% mortality at 12 months.

National practice

- Because of poor national results, the Health Care Finance Administration (HCFA) halted reimbursement for LVRS until a 'definitive' study published.
- The Center for Medicare and Medicaid Services (CMS, formerly HCFA) and the National Heart, Lung and Blood Institute (NHLBI) developed the National Emphysema Treatment Trial (NETT) (limited to 17 centers).

National Emphysema Treatment Trial (NETT)

Study design

- Prospective randomized clinical trial comparing LVRS with medical management of emphysema in patients with Medicare, or private insurance that would pay for therapy.

- Presence of bilateral emphysema with severe airflow obstruction, hyperinflation on imaging, and the ability to participate in an exercise program.
- Optimal medical therapy: smoking cessation, bronchodilators, O_2 therapy, pulmonary rehab, pneumococcal and influenza vaccines.
- Surgery: LVRS via a sternotomy, VATS, or thoracotomy.
- All patients had 6–8 weeks of pulmonary rehab prerandomization.
- Patients randomized to medical therapy received pulmonary rehab for 8–9 weeks postrandomization.

Initial results

Initial results identified a high-risk group that had higher mortality after surgery compared to medical therapy and these were removed from further analysis: [3]

- Patients with FEV_1 <20% and homogenous emphysema.
- Patients with FEV_1 <20% and DLCO <20%.

Longer-term results [4,5]

- Short- and long-term survival advantage for surgery in patients with upper lobe emphysema and low exercise capacity.
- Survival advantage for medical therapy for patients with non-upper-lobe emphysema and high exercise capacity.
- Improvements in both exercise capacity and quality of life as a result of surgery in patients with upper lobe emphysema and both low or high exercise capacity.
- Improvements in quality of life only (no survival advantage) as a result of surgery for patients with upper lobe emphysema and high exercise capacity, and patients with non-upper lobe emphysema and low baseline exercise capacity.

Impact of NETT on Medicare coverage

- Non-coverage of LVRS was extended for high-risk patients.
- Coverage for LVRS was provided for upper lobe emphysema and low or high exercise capacity for survival and palliative benefit respectively.
- Coverage for LVRS was extended for non-upper-lobe emphysema and low exercise capacity for palliative benefit.
- Types of facilities that are allowed Medicare coverage of LVRS:
  - NETT approved research centers.
  - Centers accredited by the Joint Commission on Accreditation of Healthcare Organizations (JCAHO) under the Disease Specific Certification Program for LVRS.
  - Centers approved for lung or heart and lung transplantation.

References

1. Brantigan OC, Mueller, F. Surgical treatment of pulmonary emphysema. *Am Surg* 1957; **23**(9): 789–804.
2. Cooper JD *et al.* Results of 150 consecutive bilateral lung volume reduction procedures in patients with severe emphysema. *J Thorac Cardiovasc Surg* 1996; **112**: 1319–30.
3. NETT Group. Patients at high risk of death after lung-volume-reduction surgery. *New Engl J Med* 2001; **345**: 1075–83.
4. NETT Group. A randomized trial comparing lung-volume–reduction surgery with medical therapy for severe emphysema. *New Engl J Med* 2003; **348**:2059–73.
5. Naunheim KS *et al.* Long-term follow-up of patients receiving lung-volume-reduction surgery versus medical therapy for severe emphysema by the National Emphysema Treatment Trial Research Group. *Ann Thorac Surg* 2006; **82**(2):431–43.

Lung volume reduction surgery

Key facts

- In the USA, Medicare will only reimburse lung volume reduction surgery (LVRS) for very specific patient types and defined surgeon and institution qualifications.
- The goal of surgery is to remove ventilated but poorly perfused lungs in patients with upper lobe emphysema and low exercise capacity (see Fig. 14.5).
- VATS approach is as effective as open techniques and is the preferred approach.
- Prolonged air leak is the most common postoperative complication and staple line buttressing is a necessity.

Open approach

Set-up

- General ET anesthesia with invasive arterial monitoring, and usually a central line.
- The patient is then positioned in the *supine position* (📖 p39).
- Median sternotomy, both mediastinal pleura are opened.

Lung resection

- The pulmonary ligaments are divided.
- The hyperinflated lung tissue can be punctured to allow for deflation.
- A clamp can be used to mark the area of resection, and a stapler is used to resect the lung tissue.
- The line of stapling is parallel and 3–4cm above the major fissure.
- Staple line buttress material should be used to avoid air leaks.
- Multiple commercially available staple line buttressing products.
- Care should be taken to avoid crossing fissures into the lower lobes.

VATS approach

Set-up

- General ET anesthesia with invasive arterial monitoring, and usually a central line.
- The patient is then positioned in the *lateral, decubitus position* (📖 p39) with spine parallel to and close to the edge of the table, using sandbags, or rolls and tape, usually with the upper arm positioned on a rest, the lower arm flexed in front of the head.
- Incisions are similar to VATS lobectomy (see 📖 p686).

Lung resection

- The bullous disease in the apex of the lung can be decompressed with cautery or scissors.
- Resection of tissue follows the same principles as the open technique.

Postoperative care and complications

- Patients should be extubated in the operating room.
- Epidural catheters should be liberally used or PCA.
- Aggressive pulmonary toilet is essential.

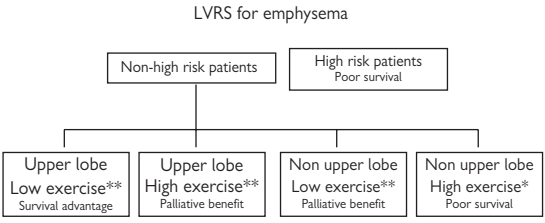


Fig. 14.5 Algorithm for surgical treatment of emphysema. ** Low and high exercise refers to exercise capacity.

- Supraventricular arrhythmias occur in 20% of patients and should be treated, though amiodarone should be avoided secondary to risk of pulmonary fibrosis.
- Bowel prophylaxis to avoid stress ulcers, and laxatives to avoid constipation secondary to narcotic use.
- 30-day hospital mortality, 3% after sternotomy and 2.1% after VATS (P=0.74).[1]

Prolonged air leak

- Most common complication is air leak.
- 54% after sternotomy and 66% after VATS.[1]
- Chest tubes should remain on water seal.
- Patients may eventually go home with Heimlich valves.

Reference

1. NETT Research Group. Safety and efficacy of median sternotomy versus video-assisted thoracic surgery for lung volume reduction surgery. *J Thorac Cardiovasc Surg* 2004; **127**:1350–60.

Pneumothorax

Key facts

- Pneumothorax (PTx) is defined as the presence of air in the pleural space with secondary lung collapse.
- Stable patients with spontaneous pneumothorax should undergo chest CT to determine underlying pathology.
- Pneumothorax is classified as: (1) primary spontaneous, (2) secondary spontaneous, and (3) post-traumatic/iatrogenic.
- Treatment is usually a pigtail chest tube: recurrent penumothoraces secondary to apical blebs may require VATS pleurodesis.

Primary spontaneous pneumothorax

Commonly seen in young, tall, male smokers. The lifetime risk of primary spontaneous PTx in smoking males is 12% vs. 0.1% in non-smoking males. More common on the right side. <10% of cases are bilateral. Usually caused by rupture of small subpleural blebs (collections of air <2cm). Usually found at the apex of the upper lobe or the apical segment of the lower lobe. The rest of the lung parenchyma is normal. May also be caused by rupture of bullae.

Presentation

- Dyspnea, chest pain, cough, tachypnea.
- Ipsilateral decreased chest wall movement, hyper-resonant hemithorax to percussion, absent breath sounds on auscultation, pleural rub, tachycardia. Rarely as tension pneumothorax (📖 p823).

Investigations

- PA CXR usually diagnostic.
- CT scan gives an accurate estimate of size of pneumothorax and is useful for assessment of remaining lung parenchyma and contralateral lung. High-resolution fine slice CT will demonstrate apical blebs.

Complications

- Tension PTx (emergency chest tube, 📖 p248).
- Pneumomediastinum.
- Haemopneumothorax.
- Recurrent PTx.

Conservative management

- Observation (unilateral, small <20% pneumothorax, no apical blebs).
- Trial of needle aspiration if minimal symptoms and moderate PTx.
- Tube thoracostomy if compromised or very large PTx.

Surgery

- Surgery is indicated in the following cases;
- *First episode:* prolonged air leak (>72h) tension PTx, hemothorax, bilateral pneumothoraces, residual collapse of lung, 100% PTx, occupational hazard, PTx secondary to giant bulla, previous contralateral pneumonectomy.
- *Recurrence of PTx:* the aim is to resect the blebs or bullae and obliterate the pleural space with adhesions either using chemical or abrasion pleurodesis or parietal pleurectomy (apical or full). It may be performed through a minithoracotomy, axillary incision, or thoracoscopically.

Recurrence rate

- <2% following surgical pleurectomy via minithoracotomy.
- 5% after VATs pleurodesis, 5–10% following chemical pleurodesis.

Secondary spontaneous pneumothorax

These patients are frequently sicker patients with primary spontaneous PTx, less likely to resolve with needle aspiration, more likely to be compromised. Formal chest tube is usually the first-line option.

Causes

- Cystic fibrosis, asthma, COPD and other bullous disease. Be wary of confusing a large bulla with a pneumothorax—inserting a chest tube could cause a tension pneumothorax. Get a CT if any doubt.
- Interstitial lung disease.
- Infections, e.g., mycobacterial, *Pneumocystis jiroveci (carinii)*, parasitic, mycotic.
- Malignancy: bronchogenic, metastatic (sarcoma and lymphoma).
- Collagen diseases, catamenial, Ehlers–Danlos syndrome, histiocytosis X, scleroderma, lymphangioleiomyomatosis, Marfan syndrome.
- Rupture of the esophagus.
- *Cystic fibrosis:* PTx found in 10% of patients. Remember these patients are possible candidates for future lung transplantation when considering management options. Full parietal pleurectomy is a contraindication to lung transplantation, but pleurodesis is not.
- *COPD:* the most common cause of secondary pneumothorax. Age usually >50 years. Patients often have very little pulmonary reserve. They may not tolerate surgical management and single-lung ventilation. Treatment options are tube thoracoscopy and chemical pleurodesis or long-term tube thoracoscopy.
- *Infection:* cavitating pulmonary lesions rupture into pleural space.
- *AIDS:* Usually secondary to *Pneumocystis jiroveci (carinii)*, and pneumonia. May be presenting feature of AIDS. Most effective treatment is surgical.
- *Catamenial:* age 20–30 years. Incidence 3–6% of women. Occurs 2–3 days following onset of menstruation. Right side more commonly affected. Usually small, presenting with dyspnea and chest pain. Pathogenesis unclear. May be a result of ectopic endometrial tissue on the diaphragm, or pleura, and possible diaphragmatic defects. Treatment is VATS resection of endometrial deposits, blebs, mechanical pleurodesis, and oral contraceptive therapy.

Primary spontaneous pneumomediastinum

- Uncommon. Males affected more frequently than females. Occurs after exertion or increased intra-abdominal pressure. Associated with cocaine, marijuana, and crack cocaine use. Caused by rupture of alveolar sacs with air tracking along peribronchial and perivascular spaces into neck.

Presentation: sudden onset chest pain, dyspnea, dysphagia, cough, subcutaneous emphysema over chest wall, Hamman's sign. CXR confirms diagnosis. ***Management:*** non-operative. Need for emergency surgical decompression rare.

Pleural effusion

Key facts

- Pleural effusions are abnormal collections of pleural fluid in the pleural space, which exceed the lymphatic drainage capacity.
- Pleural effusions are determined by Light's criteria to be transudative or exudative.

Exudate versus transudate

- ***Exudate:*** specific gravity is >1.02, protein content >2.0g/dL, and at least one of the following: ratio of pleural fluid protein to serum >0.5, ratio of pleural fluid LDH and serum >0.6, total pleural fluid LDH >2/3 upper limit of normal, ratio of pleural fluid albumin and blood <1.2.
- Other lab results that are not part of ***Light's criteria***, but are suggestive of an exudative effusion are glucose <60 mg/dL, leukocyte count >10 000/mL, pH <7.20, and elevated amylase.
- ***Transudates*** are generally 'simple' effusions that result from capillary leak syndromes, or increases in hydrostatic or oncotic pressure. Examples of transudative effusions include congestive heart failure, cirrhosis of the liver, nephrotic syndrome, sarcoidosis, and increased pulmonary hydrostatic pressure from pulmonary emboli.
- Examples of exudates include malignant effusions, chylothorax, hemothorax, empyema, pancreatitis, esophageal leak, collagen vascular disorders, autoimmune disorders, uremia, and postmyocardial infarction syndrome.
- Transudates are caused by cirrhosis of the liver, nephrotic syndrome, glomerulonephritis, congestive heart failure, myxedema, sarcoidosis, multiple pulmonary emboli.

Clinical presentation

- Small effusions are often asymptomatic. Larger effusions cause cough, chest pain, dyspnea.
- Decreased ipsilateral chest expansion, dullness to percussion, decreased breath sounds over the effusion on auscultation, crepitations may be heard.

Investigations

- ***CXR:*** small effusions are demonstrated by blunting of the costodiaphragmatic angles, larger effusions produce a fluid level with a meniscus. Decubitus films can determine free-flowing fluid versus loculated fluid. Blunting of the costophrenic angle on a PA view often represents at least 400mL of fluid.
- ***US scan:*** useful for loculated effusions, helps to localize optimal site for chest drainage.
- ***CT scan:*** useful when looking at underlying lung and pleural lesions. Gold standard in identifying fluid loculations, entrapped lung, and surgical planning.
- ***Pleural aspiration cytology:*** may obtain diagnostic information, helpful when planning treatment. Can provide Light's criteria information as well as other analysis such as Gram's stain.

Management

- Chest tube placement, either small bore or large bore (28F). May not be completely successful in loculated effusions.
- VATS evacuation of pleural fluid ± decortication, ± pleurodesis (see 📖 p734).
- For effusions that have been present >2 weeks or are associated with chest wall tenderness, or have a calcific peel or rind, VATS may not allow for successful lung expansion.[1]
- During VATS exploration, a pleural biopsy should be performed to evaluate for cause of effusion.
- Thoracotomy with decortication (see 📖 p706).

Malignant pleural effusion

- Malignant pleural effusions are exudative and are the result of tumor invading pleural lymphatics and reducing the absorptive capacity of the pleural space.
- Pleural effusions as the result of lymphoma may respond to chemotherapy, but most malignant pleural effusions have a poor prognosis.
- Symptoms include insidious progressive dyspnea, with eventual mediastinal shift.
- Only 20% of thoracentesis yield diagnostic cytology.
- Malignant pleural effusions are often bloody.
- Treatment: thoracentesis should be done initially to determine if the lung expands or not. 98% of malignant effusions recur within 30 days.
- If the lung expands, then a VATS evacuation and pleurodesis can be successful.
- VATS also provides pleural tissue to confirm tumor deposits.

Complications

- Supper infections causing empyema.
- Treatment failure with recurrence of pleural effusion.
- Damage to underlying lung parenchyma, resulting in prolonged air leak.

Reference

1. Luh SP *et al.* Video-assisted thoracoscopic surgery in the treatment of complicated parapneumonic effusions or empyemas: outcome of 234 patients. *Chest* 2005; **127**(4): 1427–32.

Pleurodesis

Key facts

- Pleurodesis is any procedure designed to cause the parietal and visceral pleura to adhere together, obliterating the pleural space with a view to preventing recurrent pneumothoraces or pleural effusions.
- This can be achieved by two mechanisms: abrasion pleurodesis (where the parietal pleura is abraded) or chemical pleurodesis (where talc or other agent is instilled into the pleural space).
- Pleurodesis can be performed via a chest tube, VATS, or open.

Procedure

Set-up

- General ET anesthesia.
- The patient in the *lateral, decubitus position* (📖 p39) using sandbags, or rolls and tape, usually with the upper arm positioned on a rest.
- Incisions are as for VATS lobectomy without the large working port.

Dealing with underlying pathology

- Evacuate any pleural effusion to dryness, sending samples for cytology (the more, the better), microbiology, and biochemistry. Inspect whole hemithorax, sending samples of abnormal pleural plaques for histology.
- If pleurodesis is being performed for pneumothorax, carefully inspect the lung parenchyma, particularly the apices of the upper lobe for any apical blebs, which should be stapled and send for histology. Blebs commonly appear in the apical or anterior segments of the upper lobes, the lateral segment of the middle lobe, and the superior segment of the lower lobes.
- If no blebs are found, the lung apex is resected prophylactically.

Pleurodesis

- Mechanical pleurodesis uses an electrocautery scratch pad or sterile scratch paper to abrade the parietal pleural of the chest wall to the point of visual tissue disruption and hyperemia.
- Chemical pleurodesis can be done by insufflating talc, docycline, or bleomycin.
- Talc is cheaper and probably more effective.
- 4–6g of aerosolized talc is insufflated into the pleural space, taking care to occlude all ports (so that it does not simply mostly escape) and carefully coat all areas of the lung and chest wall. Mechanical pleurodesis does not prevent additional blebs from forming and subsequent spontaneous PTx but may prevent complete lung collapse and tension pneumothorax.
- Talc pleurodesis are avoided in young patients, as the long-term side effects of talc are unclear, and talc pleurodesis leads to false-positive PET activity in the pleura, which may confound potential cancer staging in the future.
- Talc may be used in elderly patients, secondary spontaneous PTx, and patients with primary spontaneous PTx who have failed initial mechanical pleurodesis.
- The lung is inflated under thoracscopic vision after chest tubes placed.

- Chest tubes remain on continuous suction for 72h, and are removed when output is <200mL for 24h.

Stage I empyema

- Talc should be avoided where coinciding pleural space infection is suspected.
- Docycline may be useful for pleurodesis for effusions where infection cannot be ruled out.
- Instil solution of 500mg of doxicycline in 50mL of saline.
- If the lung does not expand, then a pleurodesis will not be effective, and a chronic indwelling pleural catheter should be placed.

Malignant pleural effusion

- An entrapped lung secondary to malignant pleural effusion should not be decorticated.
- There are commercially available indwelling pleural catheter kits.
- These catheters can be placed in the OR or at the bedside.
- Patients can drain their own pleural space when symptomatic.
- Often the catheters can eventually be removed if no need for drainage for 2 weeks.
- Complications include infection, discomfort, purchasing of catheter and bottle system.

Bedside (chest tube) pleurodesis

- If the patient is a poor surgical candidate, then a pleurodesis can be done at the bedside by inserting pleurodesing agent through a large-bore chest tube (24–28F).
- 4–6g of talc is suspended in 120mL saline to form a slurry, and inserted into the chest tube.
- The chest tube is clamped for 1h, and the patient undergoes position changes every 15min: supine left lateral decubitus, right lateral decubitus, and reverse Trendelenburg.
- The tube is unclamped, and the patient is placed on continuous suction for 72h. The chest tube is removed if total output is <200mL for 24h.

Outcomes

- VATS talc pleurodesis is 95% effective, bedside pleurodesis 75%.
- Pain is common secondary to inflammation of the parietal pleura.
- Often talc slurries can be mixed with 1% lidocaine to limit the pain. Patients may need a PCA. NSAIDs such as ibuprofen and ketorolac should be avoided as they may decrease inflammatory response and subsequently limit an effective pleurodesis.
- Fever caused by local inflammation, treated with acetaminophen.
- SIRS due to over-administration of talc: pulmonary insufficiency and hypotension.
- Two or more of the following: tachypnea >20 breaths/min, tachycardia >90bpm, temperature >38°C or <36°C, or WBC >12 000 or <4000.
- Limiting the amount of talc used to ≤ 6g, and not performing bilateral talc pleurodesis during the same setting.

Pleurectomy

Key facts

- Removal of the parietal pleura is another way of getting the visceral pleura to adhere completely to the chest wall, obliterating the pleural space and preventing recurrent pneumothorax.
- It is usually performed via VATS or a minithoracotomy in the axilla or posterolaterally

Pleural anatomy

- The pleura consists of mesothelial cells forming the visceral pleura which lines the outer aspects of the lung and the parietal pleura which lines the chest wall, mediastinum, pericardium, diaphragm, and other extrapulmonary organs of the chest cavity.
- The potential space between the visceral and parietal pleura contains pleural fluid, which lubricates the space, and allows for easy movement of the lung.
- The amount of pleural fluid produced is 0.01mL/kg/h or 200mL/24h.
- Pleural fluid is reabsorbed via the pulmonary lymphatics for the visceral pleura and direct lymphatic channels permeating the parietal pleura.
- Venous drainage is via the peribronchial veins.
- The parietal pleura contains afferent nerve fibers which can transmit pain signals, and is the layer that should be anesthetized during chest tube placement.

Procedure

Set-up

- General ET anesthesia with lung isolation if VATS.
- The patient is positioned in the *lateral, decubitus position* (📖 p39) with spine parallel to and close to the edge of the table, using sandbags, or rolls and tape, usually with the upper arm positioned on a rest, the lower arm flexed in-front of the head.

Procedure

- The aim is removal of the parietal pleura from the 4th intercostal space up to the apex, as well as addressing likely causes which are usually apical blebs which should be stapled first as they are easiest to see at the beginning of the case.
- The parietal pleura is then peeled off the chest wall either starting at the minithoracotomy incision or at the port site opposite the camera in a VATS.
- Once you have a lip of pleura big enough to grab with a tonsil clamp or a Roberts, then a gentle winding action wraps the pleura around the clamp, peeling it off the chest wall.
- An apical and basal chest tube are replaced and the lung carefully checked to make sure it re-expands.

Closure

Insufflation with 10–20mL 0.25% bupivacaine helps postoperative pain, along with the usual costal block.

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Chylothorax

Key facts

- Chylothorax is the abnormal collection of lymphatic fluid in the pleural space, usually as a result of disruption of the thoracic duct or one of its tributaries.
- It usually occurs after inadvertent damage to the thoracic duct or its tributaries during mediastinal dissection, rapidly accumulating in chest tubes as a milky fluid once the patient resumes normal diet, or forming a pleural collection.
- Pleural fluid triglycerides >110mg/dL and pleural fluid to cholesterol level ratio <1 is diagnostic.
- It usually resolves with adequate pleural drainage and a low-triglyceride diet, but occasionally surgical repair is required.

Pathology

- The thoracic duct is a valved lymph channel that originates in the abdomen as the ***cisterna chyli***, forming from vessels draining the lower limbs and vessels, entering the right thoracic cavity via the esophageal hiatus at T10–T11, travelling cephalad between the spine, descending aorta, esophagus, and azygos vein, crossing over to the left side at T4–T6. It drains into the left subclavian or jugular vein above the clavicle.
- 1.5–2.5L of chyle is produced per day, and may decrease or increase depending on the level of fat in the diet.
- Aetiology includes: trauma, iatrogenic surgical injury, malignancy, tuberculosis, mediastinitis, pancreatitis, SVC, left jugular or subclavian vein thrombosis, idiopathic, and congenital thoracic duct atresia or fistula.
- Chyle leaks have been described after almost all thoracic procedures, including left subclavian central venous line placement, but most commonly after procedures involving extensive posterior mediastinal dissection at the base of the heart, or around the esophagus.
- Spontaneous chyle leaks in adults are almost always associated with malignancy, and are secondary to obstruction of the lymphactics.

Presentation and investigation

- Diagnosis is often clinical, based on clinical suspicion and the presence of milky white fluid (pink if mixed with blood postoperatively).
- Patients who are nil by mouth may exhibit only straw-colored chyle fluid.
- Pleural fluid triglycerides >110 mg/dL and pleural fluid to cholesterol level ratio <1 is diagnostic.

Management

- Initial management is medical: nil by mouth, drainage of pleural space with chest tubes, parental nutrition, and/or medium-chain triglyceride diet.
- Octreotide (80mg, sc, 3 times daily) may reduce lymphatic flow.
- Some reported success of the percutaneous thoracic duct embolization, however this is a selective skill set that that not many interventional radiologists possess.
- Pleuroperitoneal shunts have been used for diffuse leaks, e.g., SVCO and lymphangiomatosis: they can be placed under local anesthetic.

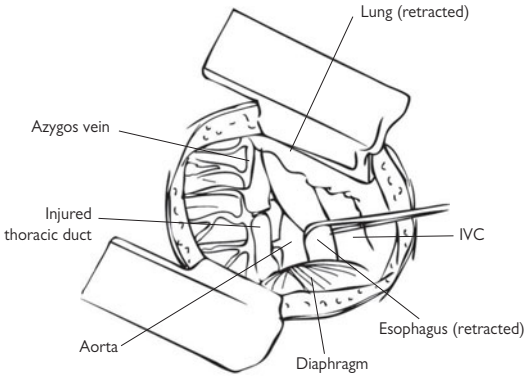


Fig. 14.6 Right thoracotomy approach to ligation of thoracic duct.

Indications for surgery

- Hypoalbuminemia, leukopenia, malnutrition secondary to chyle loss.
- 1–2L loss per day, for 1–2 weeks.

Surgical repair of thoracic duct

- Goal of surgery is thoracic duct ligation ± talc pleurodesis.
- May be done by VATS or open thoracotomy on the side of the chylothorax (or in the case of bilateral chylothorax starting on the right).
- Prior to surgery, patient is fed a high-fat diet, 6h before surgery, via a nasogastric or feeding tube, the patient is fed 80–100mL of cream per hour. This allows for marked expression of milky fluid from the disrupted thoracic duct, and facilitates identification of the injury.

Set-up

- General ET anesthesia.
- The patient in the *lateral, decubitus position* (📖 p39) using sandbags, or rolls and tape, usually with the upper arm positioned on a rest.
- Posterloateral thoracotomy in 7th or 8th intercostal space, unless previous incision in which case reopen this.
- If a specific leak is identified this is ligated with non absorbable sutures, and the thoracic duct is also ligated as it enters the thorax through the aortic hiatus by retracting the esophagus anteriorly (Fig. 14.6).
- If the duct cannot be specifically clipped or ligated, then a mass ligation where the duct should be is performed near the diaphragmatic hiatus. This involves opening the parietal pleural overlying the descending aorta, mobilizing the distal esophagus, and mass ligating all tissue between the azygos vein, esophagus, aorta. and spine.

Results of surgery

Resolution of chyle leak after surgical ligation in 85% of patients.

Pleural tumors

Key facts

- Mesothelioma is a rare and highly malignant tumor, usually unilateral, of mesothelial cells of the visceral and parietal pleura. The median survival after diagnosis is 12 months.
- The majority (80%) of cases is caused by asbestos exposure, with a latency period of at least 20 years, incidence is rising. Often seen in the 6th to 7th decade of life.
- Other pleural masses include benign pleural plaques or asbestosis (which have no potential for deteriorating to mesothelioma); metastatic tumors to pleura, e.g., thymoma, NSCLC, and fibrous tumors.

Mesothelioma

Pathology

- Male to female ratio of 5:1.
- There is no direct link between smoking and mesothelioma.
- The majority (80%) of cases are caused by asbestos exposure, with a latency period of at least 20 years, incidence is rising due to industrial exposure in previous decades. Two types of asbestos fiber have different risk of causing mesothelioma:
  - *Serpentine fibers* are large, and have poor tissue penetration, so lower risk of causing mesothelioma.
  - *Amphibiole fibers* are small, and travel through the bronchioles to the pleural surface, associated with higher risk of mesothelioma.
- Other risk factors include radiation exposure, liquid paraffin, rubber, copper, fiberglass, nickel, aerosolized glass, industrial mineral oil, tuberculosis, pleuritis and other chronic pleural infections, hereditary, familial Mediterranean fever.
- At-risk individuals include construction workers, mechanics, pipe fitters, military personnel with contact with pre-Vietnam era ships and planes, and family members of all of these individuals who would have been exposed to their clothing,
- *Histology:* vimentin –ve and cytokeratin +ve. Three tumor types:
  - Epithelial: commonest histology and most favorable prognosis.
  - Sarcomatoid: least favorable prognosis.
  - Mixed epithelial and sarcomatoid: also with poorer prognosis than pure epithelial.
- Tumor usually unilateral, (although majority of patients have occult metastases at death—abdomen, contralateral lung). Progresses locally encasing lung, and invading chest wall, diaphragm, and mediastinum.

Presentation

May be asymptomatic. Patients usually present in their 6th or 7th decade with fatigue, chest pain, dyspnea, weight loss, cough, fever, weakness, anorexia (Horner's syndrome, hemoptysis, spontaneous pneumothorax, dysphagia). *Examination:* pleural effusions are an early finding. Diminished lung expansion on affected side. Dullness to percussion. Decreased breath

sounds on auscultation. Much less commonly palpable chest wall mass, palpable supraclavicular lymph nodes.

Investigation

- CXR may demonstrate pleural thickening and any pleural effusion.
- CT scan helpful when staging tumor and monitoring response to treatment: PET CT to identify distant metastases.
- MRI useful for detection of mediastinal and brain involvement.
- Pleural biopsy to obtain histological diagnosis (tumor seeding of incision common: liaise with surgeons/oncologist to plan site).
- Pleural fluid cytology is positive in up to half of patients.

Treatment

- Primarily palliative: VATS talc pleurodesis (📖 p734) can ameliorate symptoms from malignant pleural effusion, and is often the route to obtaining definitive histological confirmation of diagnosis.
- Pleurodesis may make subsequent pleurectomy easier.
- Curative intent: multimodality therapy combining radiation and often doxorubicin-based chemotherapy with surgery (extrapleural pneumonectomy 📖 p745) is the therapeutic approach of choice.

Outcomes

Median survival from diagnosis is 12–18 months.

Fibrous pleural tumors

Pathology

- Rare tumors that tend to occur in the 6th decade of life; may be benign (80%) or malignant and have no association with asbestos exposure.
- Originate from the mesenchymal layer beneath the mesothelial layer.
- Men and women are affected equally.
- May be pedunculated or sessile.
- *Histology:* vimentin +ve and cytokeratin –ve:
  - Benign: arise from parietal or visceral pleura, though more commonly the visceral pleura. Usually small. Few mitosis with <4 per high powered field (HPF).
  - Malignant: often sessile and non-pedunculated, characterized by increased cellularity and >4 mitosis per HPF, with pleomorphism.

Presentation and investigation

Often asymptomatic, dyspnea, pleural effusion, pleuritic chest pain, hypertrophic pulmonary osteoarthropathy (clubbing), and hypoglycemia from tumor secreting insulin-like growth factors. CXR and chest CT.

Treatment

- Pedunculated lesions may require only VATS local resection.
- Sessile and large lesions may require VATS or open resection with associated lobectomy or *en bloc* resection of adjacent tissue such as the chest wall.

Outcomes

- Benign tumors with R0 resection have a <10% recurrence rate.
- Malignant tumors with R0 resection have a 14–63% recurrence rate and 30% mortality.

Staging mesothelioma

Key facts

- Staging based on thoracic surgical anatomic principles.[1]
- Staged using TNM tumor classification, but various other staging systems exist.

Stages

- Stage I: disease confined within the capsule of the parietal pleura (including the ipsilateral pleura, lung, pericardium, and diaphragm).
- Stage II: all of stage I with positive N1 and/or N2 lymph nodes.
- Stage III: local extension into the chest wall, mediastinum, heart, through the diaphragm or peritoneum, but no N3 or distal node involvement.
- Stage IV: distant metastatic disease.
- Staging based on the International Mesothelioma Interest Group.[2]

Tumor stage

- T1a: tumor limited to ipsilateral parietal pleura with no involvement of the visceral pleura.
- T1b: tumor limited to ipsilateral parietal pleura with involvement of the visceral pleura.
- T2: tumor involving all of the ipsilateral pleural surfaces with at least one of the following features: Extension from visceral pleura into the underlying pulmonary parenchyma or involvement of the diaphragm.
- T3: locally advanced but resectable tumor. Tumor involving all of the ipsilateral pleural surfaces with at least one of the following features: extension into mediastinal fat, involvement of the endothoracic fascia, non-transmural involvement of the pericardium or solitary foci of tumor extending to the soft tissues of the chest wall.
- T4: locally advanced and technically unresectable tumor. Tumor involves all of the ipsilateral pleural surfaces with at least one of the following features: direct extension to mediastinal organs, contralateral pleura, spine, or peritoneum.

Nodal stage

- Nx: lymph nodes cannot be assessed.
- N0: no regional lymph node metastasis.
- N1: ipsilateral hilar or peribronchial lymph nodes.
- N2: ipsilateral mediastinal lymph nodes.
- N3: contralateral mediastinal lymph nodes, or ipsilateral or contralateral supraclavicular lymph nodes.

Metastases

- Mx: distant metastases cannot be assessed.
- M0: no distant metastases.
- M1: distant metastases.

Staging

- Stage Ia: T1aN0M0.
- Stage IB: T1bN0M0.
- Stage II: T2N0M0.
- Stage III: T3N0M0 or any T and N1 or N2.
- Stage IV: T4N0M0, or any TN3M0 or any T, any N and M1.

Staging for possible curative resection

Curative resection, which is invariably combined with multimodal therapy, carries substantial morbidity, and patients who will benefit from this strategy must not have metastatic disease, and must be able to tolerate not only pneumonectomy, but also the additional respiratory embarrassment caused by diaphragmatic resection. Routine staging (in addition to a detailed cardiopulmonary assessment) of patients being considered for extrapleural pneumonectomy includes:

- MRI brain and chest (to identify metastatic spread and local mediastinal invasion).
- CT of chest and abdomen with contrast with PET scan.
- Bronchoscopy and EGD.
- Staging laparoscopy with peritoneal lavage.
- Mediastinoscopy.

References

1. Sugarbaker DJ *et al.* Node status has prognostic significance in the multimodality therapy of diffuse, malignant mesothelioma. *J Clin Oncol* 1993; **11**(6):1172–8.
2. Rusch VW. A proposed new international TNM staging system for malignant pleural mesothelioma. From the International Mesothelioma Interest Group. *Chest* 1995; **108**(4):1122–8.

Treatment of mesothelioma

Key facts

- ***Multimodality therapy*** combining radiation and often doxorubicin-based chemotherapy with surgery is the therapeutic approach of choice: there is almost no role for surgery in isolation.
- At presentation the majority of patients are candidates only for palliation: VATS talc pleurodesis palliates malignant pleural effusion.
- ***Tumor seeding*** means that surgical biopsy and pleurodesis, even in palliative patients, should be planned with the involvement of the oncologist and the surgeon likely to perform pneumonectomy, as the port site should be where it can readily be excised as part of the thoractomy incision, or irradiated without causing radiation pneumonitis.
- ***Extrapleural pneumonectomy***, and occasionally ***lung-sparing tumor resection*** are the two options for surgery with curative intent: choice is partially dictated by planned adjuvant therapy—if radiation is planned then pneumonectomy must be performed to avoid radiation pneumonitis.

VATS biopsy and pleurodesis for mesothelioma

- This differs from VATS biopsy and pleurodesis for other conditions, because of the high incidence of ***tumor seeding*** along incision and biopsy sites.
- A single-port technique is used.

Procedure

- In advanced cases where extensive pleural plaque is shown on CT but large effusion is not a component, it is often possible and desirable to obtain a biopsy of the parietal plaque without entering the pleural space to reduce the risk of air leak and infection.
  - A small skin incision (10mm in thin individuals 3–4cm in obese patients) is made based on where the CT scan indicates extensive parietal involvement, and the tissue carefully separated until the thick white rind is encountered.
  - This is biopsied taking care not to breach the pleural space, and checked by irrigating the wound with saline and ventilating the lung: the saline should not disappear, and there should be no air bubbles.
  - This is closed in layers with absorbable sutures.
- Where the primary goal is biopsy of less advanced disease or pleurodesis, a 15mm port site is created at a suitable location.
  - A 30° angled scope is introduced carefully into the pleural space.
  - A biopsy forceps is introduced using the same incision.
  - Avoid biopsying visceral lesions to minimize the risk of an air leak.
  - Take multiple biopsies from multiple sites.
  - After insufflating talc inflate the lung under direct vision, ensuring it expands as fully as possible without air leaks, and close the incision in layers carefully, with absorbable sutures.

Extrapleural pneumonectomy

Set-up

- General ET anesthesia, bronchoscopy (contralateral endobronchial mesothelioma is a contraindication to curative resection).
- Place NG tube (enables esophagus to be readily palpated).
- Reposition the patient in the *lateral, decubitus position* (📖 p39) using sandbags, or rolls and tape, and recheck lung isolation.
- Posterolateral thoracotomy in 7th or intercostal space, excising previous biopsy sites as full thickness ellipses down to muscle, trying to keep serratus intact, and excising 7th rib in subperiosteal fashion.

Extrapleural pneumonectomy

- The plane between the 8th rib and the parietal pleura is developed with a combination of cautery, scissors, and finger dissection, and the same is done with the 6th rib so that a retractor can be inserted.
- Dissection results in diffuse bleeding which can be cauterized using argon beam or spray settings on standard electrocautery.
- Structures to pay particular attention to include the azygos, esophagus, and IVC on the right, and the descending aorta on the left.
- The lung and tumor mass is bulky, usually making it difficult to complete the hilar dissection before the lung is completely mobilized.
- Once the tumor is completely separated from the bony chest wall, it is dissected off the pericardium which occasionally offers a reasonable plane. Pericardial involvement mandates resection: the pericardium is repaired using a Gore-Tex patch ensuring adequate 'give' to prevent hemodynamic compromise, and incised to prevent tamponade.
- The diaphragm is preserved is possible: if a small resection is performed the defect is closed primarily. Large defects are patched, taking care to preserve the phrenic nerve.
- The pulmonary artery and veins are divided, and the bronchus is divided flush with the carina, full mediastinal lymph node dissection is carried out, on right-sided resections the thoracic duct is ligated.
- Meticulous care is taken closing the wound.

Outcomes

- Operative mortality is 4–10% depending on patient selection.
- Mesothelioma generally carries a poor prognosis regardless of therapy; some studies have reported up to 50% survival in the most favorable of clinical characteristics, but 10-year survival is near 0%.
- Intraoperative instilled intrathoracic heated cisplatin immediately after extrapleural pneumonectomy has been described.

Diaphragmatic hernia

Key facts

- These hernias may be congenital hernia defects in the diaphragm that occur at anatomic points of weakness (Fig. 14.7). With time, these points of weakness allow the herniation of abdominal contents into the mediastinum or hemithorax.
- Acquired hernias can be sliding esophageal, or paraesophageal hernias (📖 p788), or occasionally due to trauma, presenting early or late after both penetrating and blunt trauma.
- Treatment is usually surgical, to prevent intestinal obstruction and respiratory compromise: repair must take into account blood and nerve supply of the diaphragm (Fig. 14.7) to preserve function.

Bochdalek hernia

- ***Pathology:*** rare hernia that occurs in 1 in 5000 births. Due to *in utero* failure of pleuro-peritoneal folds to join, resulting in a left-sided posterior defect (Fig. 14.7). 1/3 have other congenital abnormalities.
- ***Presentation:*** often asymptomatic. If occurs *in utero*, may develop pulmonary hypoplasia: these infants may be critically ill. In adulthood dyspnea, atypical chest pain, tachycardia, bowel obstruction.
- ***Investigation:*** ultrasound, CXR, and chest CT.
- ***Management:*** surgical therapy is recommended even if asymptomatic, although timing challenging as infants may require ECMO support.
  - Reduction of hernia contents and primary closure usually accomplished via transverse incision 3cm below costal margin.
  - Identify and resect the hernia sac to prevent cyst formation.
  - Defect can be closed primarily if small or closed with mesh.
  - Viscera may need to be accommodated in prosthetic silo.
  - Mortality 12–50%.

Hernia of Morgagni

- ***Pathology:*** represents only 5% of the congenital defects of the diaphragm. Occurs in the space of Larrey between the xiphoid process and distal body of the sternum and the costochondral attachments of the diaphragm (Fig. 14.7) where IMA passes through diaphragm to become superior epigastric artery. More commonly occurs on the right side. Predilection towards females. Rare in children.
- ***Presentation:*** often asymptomatic, but can have symptoms of dyspnea, bowel obstruction, and atypical chest pain.
- ***Investigation:*** CXR and chest CT.
- ***Management:*** surgical repair. Usually via subcostal or midline laparotomy, or laparoscopy. Principles include reduction of hernia contents, resection of the hernia sac, and reinforcement of the hernia repair with mesh if of significant size.

Traumatic diaphragmatic hernias

- ***Pathology***: occasional complication of penetrating or blunt trauma. Usually stomach or small intestine herniates. Predominantly left-sided.

- ***Presentation:*** often asymptomatic, but can have symptoms of dyspnea, bowel obstruction, and atypical chest pain. May present months or years after trauma with symptoms of intestinal obstruction.
- ***Investigation:*** CXR and chest CT.
- ***Management:*** surgical repair. Usually via posterolateral thoracotomy. . Principles include reduction of hernia contents, resection of the hernia sac, and reinforcement of the hernia repair with mesh.

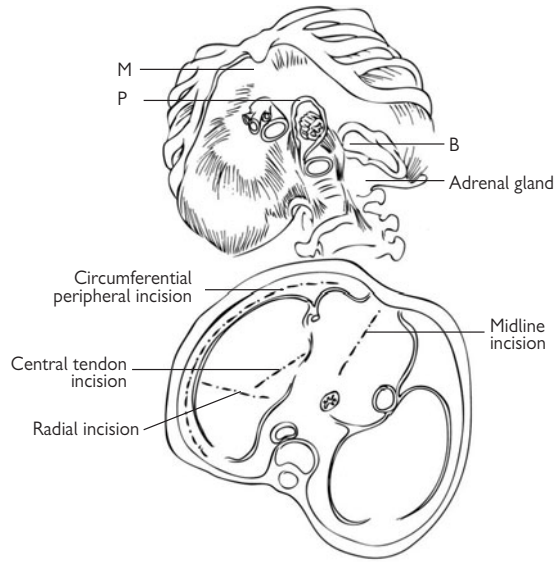


Fig. 14.7 Top image shows location of most common diaphragmatic hernias M (Foramen of Morgagni), B (Bochdalek), P (paraesophageal). Bottom image shows the relationship of phrenic nerve branches to the most commonly used diaphragmatic incisions.

Diaphragmatic paralysis

Key facts

- Congenital (***eventration***) or acquired, where the hemidiaphragm does not contract appropriately.
- This results in paradoxical motion of the diaphragm with respiration, reducing RV, TLC, and VC, and causing respiratory insufficiency.
- Acquired diaphragmatic paralysis is most commonly due to phrenic nerve injury during cardiothoracic surgery (often transient due to ice in pericardial sling, rather than permanent injury from inadvertent division), or to spinal cord injuries at or above C4–5.
- For symptomatic patients the primary therapy is surgical plication.

Pathology

- ***Acquired diaphragmatic paralysis*** is often a result of injury to the right or left phrenic nerves, which provide efferent innervation to the diaphragm. The phrenic nerve is derived from the C3–C5 cervical nerve roots. As a result, a spinal cord or cervical injury above those nerve roots can also cause diaphragmatic paralysis.
- Transient, usually unilateral, diaphragmatic paresis is seen after cardiac surgery, and attributed to ***cold injury*** from slush in the pericardial well. This can be minimized by use of insulating pads. Traction injuries, and direct injuries (during thymectomy, cardiac surgery particularly pericardiectomy, resection of mediastinal mass, pericardial window) are less common causes.
- ***Congenital eventration of the diaphragm*** is a result of a defect in the central part of the diaphragm during development, resulting in a thin membranous appearance of either one side or bilateral. The phrenic nerve is unaffected. Congenital eventration is often associated with other congenital abnormalities.
- Other causes of diaphragmatic paralysis includes viral illness (Guillain–Barré), trauma, idiopathic.
- Diaphragmatic paralysis results in ***paradoxical motion of the diaphragm*** (upwards on inspiration, downwards on expiration), reducing RV, TLC, and VC, and causing respiratory insufficiency.

Presentation and investigation

- Dyspnea, orthopnea, air hunger. Failure to wean mechanical ventilation.
- CXR shows ipsilateral elevation of the hemidiaphragm in unilateral paralysis, bilateral paresis may be less obvious.
- Fluoroscopy (***sniff test***): documents paradoxical motion of the diaphragm as the patient sniffs quickly, or breathes in, the unaffected diaphragm displaces caudally, while the ipsilateral diaphragm displaces cephalad paradoxically, causing the reduction of ipsilateral lung capacity, and movement of the mediastinum to the contralateral side.
- US: similar principle as the fluoroscopic sniff test, documenting paradoxical motion of the ipsilateral diaphragm with inspiration.
- Pulmonary function tests: spirometry is obtained to have a baseline values, and document improvement post surgical therapy. Spirometry values themselves are not diagnostic.

Management

- Primary therapy is surgery.
- For patients with phrenic nerve injury, surgical therapy of choice is ***diaphragm plication.***
- For patients with high cervical lesions (i.e., quadriplegics with lesions above C3, and intact phrenic nerve) ***diaphragmatic pacing*** may enable patients to wean off of mechanical ventilation.

Indications for surgery

- Presence of symptoms.
- Documented paradoxical motion of the diaphragm.
- No improvement after 6 months.
- In postsurgical or idiopathic etiology, the phrenic nerve may be paretic and not completely disrupted, and may recover with time.

Diaphragmatic plication

- The aim of plication is to immobilize the hemidiaphragm, preventing it from moving in a paradoxical manner.
- Can be performed via posterolateral thoracotomy (6th intercostal space), VATS, or laparoscopically.
- Using a running U stitch (0 Ethibond) and felt pledgets to limit injury to the diaphragmatic muscle, several rows are placed, and when tied, the diaphragm shortens (plicates; Fig. 14.8).
- Care must be taken to grasp and elevated the diaphragm with Allis clamps, to avoid suturing bowel on the other side of the diaphragm (generally not an issue on the right given the presence of the liver).

Diaphragmatic pacing

- Performed in patients with spinal cord injuries, to help wean mechanical ventilation. Pacing electrodes can be placed in the intact phrenic nerve in the neck, chest via VATS or directly into the diaphragm laparoscopically.
- A pacing transmitter controls respiratory rate and duration of inspiration.

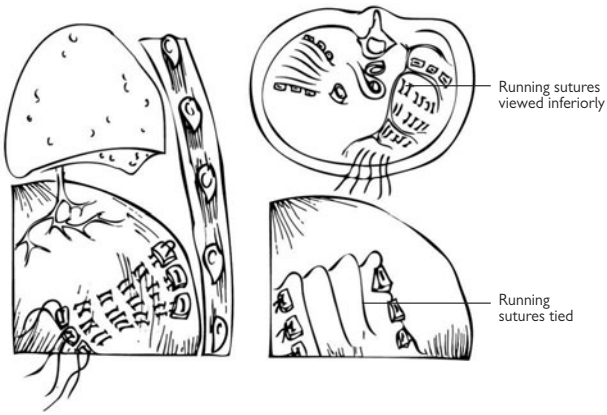


Fig. 14.8 Diaphragmatic plication.

Tracheomalacia and stenosis

Key facts

- ***Tracheomalacia*** is characterized by progressive weakening of the walls of the trachea, leading to dynamic (>90%) collapse of the airway with expiration.
- ***Tracheal stenosis*** is narrowing due congenital or acquired disorders, commonly pressure injury from ET intubation or previous tracheostomy.

Tracheomalacia

Pathology

- May be due to a congenital structural abnormality in the tracheal wall. Associated with esophageal atresia, tracheoesophageal fistula, and double aortic arch. It may occur in isolation.
- Acquired tracheomalacia may occur as a result of trauma, intubation injury, COPD, extrinsic compression of the trachea (e.g., thyroid goiters) and mediastinal tumors.

Presentation

Symptoms are due to respiratory airways obstruction. Barking cough and recurrent pneumonia are common in children. Most severe cases are characterized by cyanosis, apnea, bradycardia and, if untreated, cardiac arrest. This particularly occurs at 2–3 months of age, during or shortly after feeding. In adults some cases of tracheomalacia are not diagnosed until there is failure of extubation following a surgical procedure: most commonly after resection of massive goiters or mediastinal masses.

Investigations

- CXR: shows a narrowing of the trachea on a lateral view.
- Bronchoscopy: confirms diagnosis. The patient must be self-ventilating and not paralyzed for full evaluation of the airways. *Awake fiberoptic flexible bronchoscopy* may reveal anterior and posterior collapse of the tracheal lumen during expiration. Coughing may cause total airway obstruction.
- Dynamic expiratory chest CT also is very sensitive. Pulmonary function tests may demonstrate an obstructive pattern, but are not very specific for tracheomalacia.

Management

Minor degrees of airway collapse in all age groups may be observed. Symptoms improve during childhood. More severe respiratory compromise in children and adults require tracheal stent insertion and possible subsequent surgical intervention.

- *Stent trial*: a silicone Y stent (📖 p753) may be placed to determine if symptoms improve. Mucus plugs are a potential complication.
- If after a 2-week trial, if the patient has marked improvement in symptoms, then they may benefit from surgical therapy from a *tracheobronchoplasty* (📖 p755).

Tracheal stenosis

Pathology

- Congenital: caused by absence of part of membranous portion of tracheal segment. Manifests as multiple, complete, and stenotic tracheal rings. Often associated with other abnormalities of the tracheobronchial tree. Children usually present within the first few months of life with stridor or recurrent pneumonias. Less severe cases may present with exertional stridor.
- Intubation injuries:
  - Prolonged ET intubation with elevated (>20mmHg) cuff pressures (usually circumferential full-thickness lesion).
  - Prior tracheostomy (usually anterior triangular shaped full thickness defect from erosion around the tracheostomy site due to constant movement of the patient relative to the ventilator tubing).
  - If tracheostomy too high, subglottic injury results.
- Chronic inflammatory conditions, e.g., Wegner's granulomatosis.
- Chronic infection such as tuberculosis.

Presentation and investigation

- Symptoms usually appear up to 6 weeks following extubation.
- Clinically significant lesions present as upper airways obstruction, with exertional dyspnea, progressing to wheeze, and finally stridor (which may herald imminent *complete airway obstruction*). At this point the residual airway lumen may be <1cm, and has the potential to be rapidly and completely obstructed by mucus, aspirate, or inflammation, including attempts to intubate.
- Spiral CT scanning with 3D reconstruction (tracheal tomogram) is the investigation of choice, MRI.
- Bronchoscopy is essential in all patients:
  - Flexible awake to evaluate vocal cord function: *do not attempt to pass lesion*—this may result in loss of the residual airway due to inflammation, secretion, or bleeding.
  - Equipment for emergency dilatation and tracheostomy should be available.

Medical management

Observation may be indicated in the very young, especially those with long segment stenosis. Results may be improved if surgery is delayed until the patient is larger.

Surgery

There are several surgical options including:

- Dilation: balloon, Savary guidewire dilation, Jackson dilation via a rigid bronchoscope—this is most use in the setting of inflammation in the region where tracheal resection is planned, and buys time for this to settle before definitive surgery.
- Rigid bronchoscopy for dilation or core-out of granulation tissue.
- Tracheoplasty (📖 p755) for congenital tracheal stenosis.
- Tracheal resection and reconstruction of the stenotic segment (📖 p755).

Tracheal tumors

Key facts

Tracheal tumors are rare.

Primary tumors

Incidence rare, <2 per million persons per year. In children most are benign (90%), in adults most are malignant (>90%), mainly squamous cell carcinoma and adenoid cystic carcinoma.

Benign primary tumors

- Epithelial origin: trapilomas (viral etiology HPV6 & 11) increased risk of malignant change in patients HPV11 +ve. Carcinoid tumor: neuroendocrine tumor. Hypervascular.
- Mesenchymal origin: chondroma (most frequent, usually found in the upper trachea), Schwannoma, fibroma, lipoma, neurilemmoma hemangioma, leiomyoma, myoblastoma, polyps.

Malignant primary tumors

Adenoid cystic carcinoma

Male to female ratio 1:1. All age groups. Not related to smoking. Slowly progressive tumors. Spread occurs both circumferentially and longitudinally. Metastatic spread most commonly to lungs, also occasionally to brain, bone, and liver.

Squamous cell carcinoma

Male to female ratio 3:1. Age range 6th and 7th decades. Strongly associated with cigarette smoking. Hemoptysis common. Lymph node metastases at time of diagnosis common.

Rare, malignant primary tracheal tumors

- Spindle cell carcinoma.
- Adenocarcinoma.
- Small-cell carcinoma.
- Chondrosarcoma.
- Leiomyosarcoma.
- Carcinosarcoma.
- Melanoma.
- Synovial cell sarcoma.
- Mucoepidermoid carcinoma.

Presentation and investigation

Clinically significant lesions present as upper airways obstruction, with exertional dyspnea, progressing to wheeze, and finally stridor (which may herald imminent ***complete airway obstruction***). At this point the residual airway lumen may be <1cm, and has the potential to be rapidly and completely obstructed by mucus, aspirate, or inflammation, including attempts to intubate. Evaluation includes bronchoscopy ± rigid bronchoscopy, endoscopic biopsy to confirm malignant or benign, neck and chest CT with 3D reconstruction of the trachea, and PET/CT if malignant for staging.

Management

- For malignant tracheal tumors, there is no data to support neoadjuvant chemoradiation.
- Endoscopic techniques such as laser, argon plasma coagulation, cryotherapy, endoluminal stenting can be considered in patients who are poor operative candidates and/or palliative intent.
- Tracheal resection and reconstruction.
- Adjuvant radiation may be considered after resection of aggressive tumors or R1 resections.

Tracheobronchial stents

Indications

Used for both benign and malignant strictures, usually unsuitable for surgical resection, either due to anatomical position or general condition of patient. Most common tracheal strictures are a result of prolonged mechanical ventilation or long-term presence of tracheostomy. Also indicated in the palliation of malignant obstruction either secondary to intraluminal tumor or extrinsic compressions.

Types of stents

Airway stents should be easy to place and replace. They are placed under bronchoscopic visualization. They should not interfere with normal respiration and should have adequate intraluminal diameter to allow clearance of secretions.

- *T-tubes:* made of silicone rubber. Generally very well tolerated. May be left *in situ* for many years. T-tubes are placed through an existing stoma, or a stoma created in the centre of a tracheal stenosis.
- *T-Y tube:* similar to T-tube but longer with Y-shaped extension for placement in to left and right main bronchi.
- *Y-stents:* ideal for patients with distal tracheal, carinal, or proximal main bronchial strictures. Various lengths available.
- *Bronchial stents:* made of silicone. Flanged on both ends to prevent dislodgement.
- *Self-expanding stents:* made from stainless steel or nitinol (nickel + titanium). Uncovered stents are useful when crossing the origin of lobar bronchi as aeration of that lobe is uninterrupted. A covered version of these stents prevents intraluminal growth of either granulation tissue or tumor. Particularly useful in palliation of malignant disease. Also, often covered stents can be temporary.

Complications

- Obstruction secondary to mucous plugging, granulation tissue, tumor, displacement, or anatomical obstruction.
- Erosion into adjacent structures.

Tracheal reconstruction

Contraindications to surgery

- Metastatic disease.
- Long segment (>5cm) of involved trachea, where an anastomosis cannot be accomplished despite releases. Usually in an adult, up to 50% of the trachea can be resected with successful reconstruction.
- Invasion of tumor into adjacent structures other than the esophagus.

Reconstruction of the upper half of the trachea

Set-up

- The patient is positioned with the neck extended. General anesthesia is obtained. A rigid bronchoscopy may need to be performed to dilate and establish an airway. The chest is prepped to allow for a partial sternotomy if needed.
- A collar incision is performed, and skin flaps are made extending to the sternocleidomastoid muscles laterally, cricoid cartilage superiorly and the sternal notch inferiorly.
- The thyroid and anterior trachea is exposed.
- The isthmus of the thyroid is divided and dissected off of the trachea. The pretracheal plane is bluntly dissected. There should be minimal use of cautery, or alternatively a bipolar cautery can be used to avoid injury to the recurrent laryngeal nerves.
- An intraoperative bronchoscopy is performed to identify the location of the stenosis or tumor, and externally a needle is placed to mark the area of the lesion. Traction sutures are placed on either side of the marked area.

Tracheal resection and reconstruction

- The affected area of trachea is resected, and the patient is ventilated using a sterile armored ET tube and cross-table ventilation.
- The traction sutures are used to bring the trachea together, and the head is flexed to relieve tension.
- The trachea is re-anastomosed or reconstructed using 3/0 Vicryl sutures as corner stitches, a 4/0 running PDS for the membranous wall, and interrupted figure-of-eight 3/0 Vicryl sutures for the anterior cartilaginous wall.

Closure

- The wound is closed in layers, no drain is needed.
- *A guardian or 'Grillo' 0 silk stitch* is placed suturing the chin to the anterior chest wall, not as strength but to remind the patient not extend their neck.

Reconstruction of the lower half of the trachea

- Lower tracheal lesions are best approached via a right posterior lateral thoracotomy through the 4th intercostal space.
- The general surgical techniques and principles are the same for surgical tracheal resection and reconstruction.

Tracheobronchoplasty

Tracheobronchoplasty reconstitutes the normal morphology D shape of the trachea in ***tracheomalacia***, and prevents collapse of the membranous wall against the cartilaginous rings.

Set-up

- General ET anesthesia, lung isolation, lateral decubitus.
- Right posterolateral thoracotomy.
- Dissection of the posterior wall.

Reconstruction

- The distal trachea, main bronchi and bronchus intermedius are measured and a Y-shaped polypropylene mesh with 0.5cm lateral overlap fashioned to size.
- Using 4/0 polypropylene interrupted, partial thickness, mattressed sutures, the membranous wall is plicated to the mesh, restoring the anatomic D shape of each ring.

Outcomes

- Perioperative mortality around 5%, with morbidity including respiratory infection (11%) and cardiac arrhythmia (8%).
- Results in significant improvement of quality of life as defined by the St. George Respiratory Questionnaire, Karnofsky performance status, and 6-minute walk test.

Releases

Generally in the adult, 4–5cm of the trachea can be resected and a reconstruction can be performed without tension. A release can be performed to provide an additional 1–2cm of mobility, e.g., ***thyrohyoid (Dedo)*** or ***suprahyoid (Montgomery)*** release for upper half reconstructions, and a ***hilar release*** for lower half reconstructions. The suprahyoid release is favored as it results in less risk to the superior laryngeal nerve.

Suprahyoid release

- Transverse incision is made over the hyoid bone, and the insertions of mylohyoid, geniohyoid, and genioglossus tendons are divided.
- The attachments of the digastric muscles are divided.
- The greater cornu of the hyoid bone are divided. Drain placed.

Hilar release

On the side of the thoracotomy, the pericardium surrounding the pulmonary hilum is incised, and the pulmonary ligament is divided.

Postoperative care and outcomes

- Ideally patients should be extubated at the end of the procedure. There should be aggressive pulmonary toilet. The patient is nil by mouth for 24–48h. A fiberoptic bronchoscopy is performed at the end of the operative procedure and prior to discharge.
- Postoperative complications include pneumonia, arrhythmia, and anastomotic breakdown, especially in patients obese and diabetic patients. Anastomotic complications can be managed with an appliance or stent.
- 1% mortality and 9% anastomotic complications in expert centers.

Foreign body inhalation

Presentation

- Infants and toddlers mainly.
- Male to female ratio 2:1.
- Children usually inhale vegetable matter or nuts whereas adults commonly inhale bones.
- Immediately following inhalation there may be violent bouts of coughing, choking, and gagging with possible complete airway obstruction.
- This is usually followed by an asymptomatic interval, where there is an absence of signs and symptoms as the foreign body becomes lodged.

Upper airway foreign bodies

Usually bone or eggshell. They become lodged between the vocal cords and usually cause complete obstruction.

Tracheal foreign bodies

Usually causes a partial obstruction. There may be an audible slap and palpatory thud.

Bronchial foreign bodies

Signs change rapidly until foreign body lodges within a bronchus. There is wheezing, coughing, and decreased ipsilateral air entry.

Complications

- Obstruction with infection distal to the foreign body.
- Cough, hemoptysis, and fever.
- Atelectasis, lung abscess, pneumonia.
- Bronchial granulation tissue, fistula formation.

Investigation

The majority of foreign bodies are radiolucent. Early radiography will demonstrate air trapping with mediastinal shift to the unaffected side. Radiopaque objects will be seen on chest radiographs. Radiographs taken a week or more after the event may demonstrate atelectasis distal to a foreign body.

Treatment

Usually not a surgical emergency. Endoscopic removal usually performed within 24h of presentation. Flexible instruments should not be used as they are unable to adequately control the airways. A rigid bronchoscope and retrieving forceps are the instruments of choice. If the foreign body is spherical a balloon may be passed deflated beyond the foreign body. It is then inflated and the balloon is withdrawn. This dislodges the object and allows easier extraction.

Postoperative course

Usually uneventful. Any evidence perioperatively of infection should be treated with antibiotics and physiotherapy. Patients are usually admitted for observation overnight.